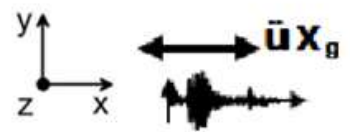
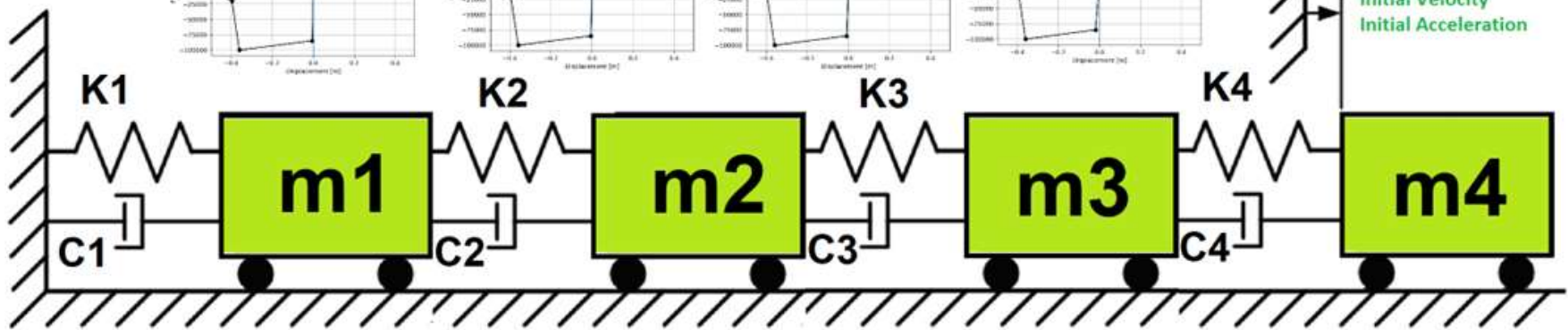
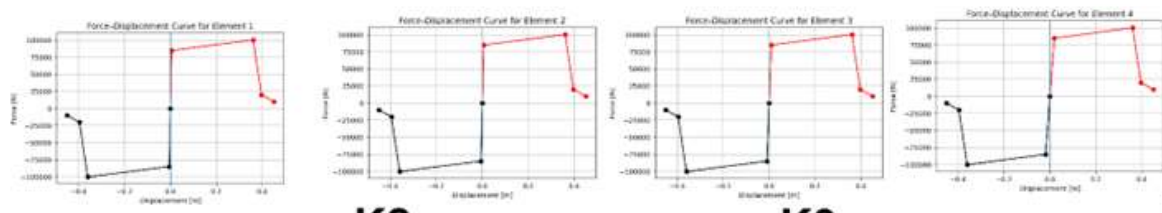


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

# **NONLINEAR DYNAMIC ANALYSIS UNDER FREE-VIBRATION COMPUTATION AND VISUALIZATION RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING OPENSEES**

**(CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM)**

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

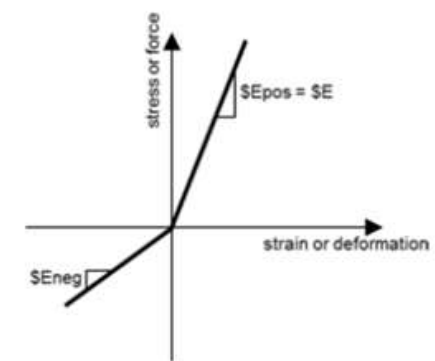
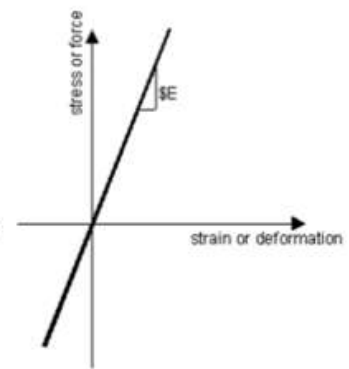


$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

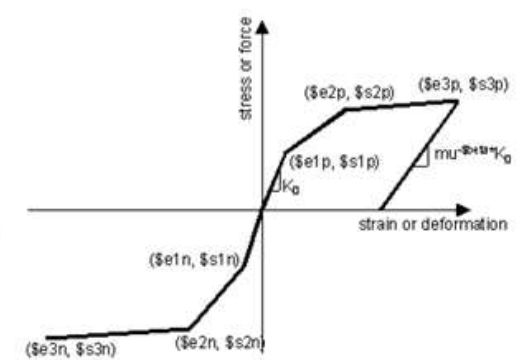
$\Delta_d$  = Lateral Displaement from Dynamic Analysis

$\Delta_y$  = Lateral Yield Displaement from Pushover Analysis

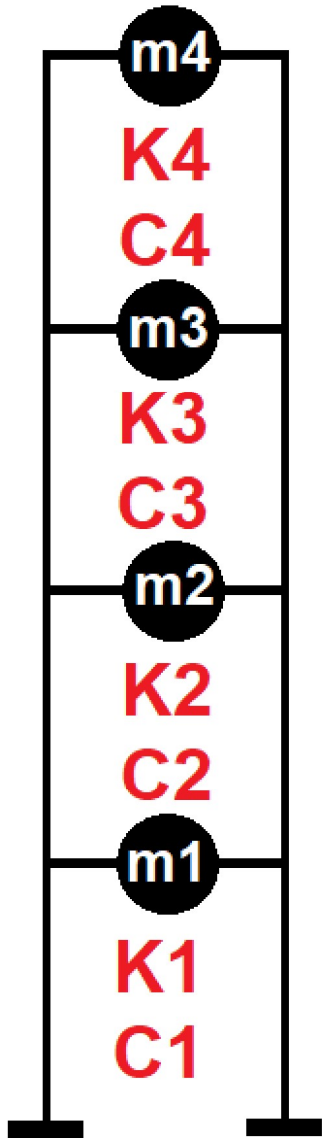
$\Delta_u$  = Lateral Ultimate Displaement from Pushover Analysis



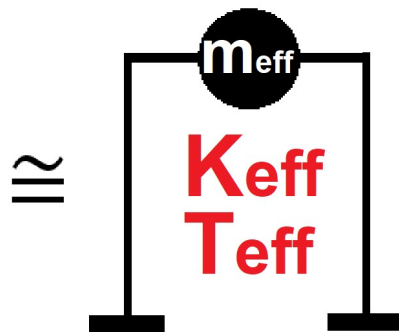
ELASTIC MATERIAL



INELASTIC MATERIAL

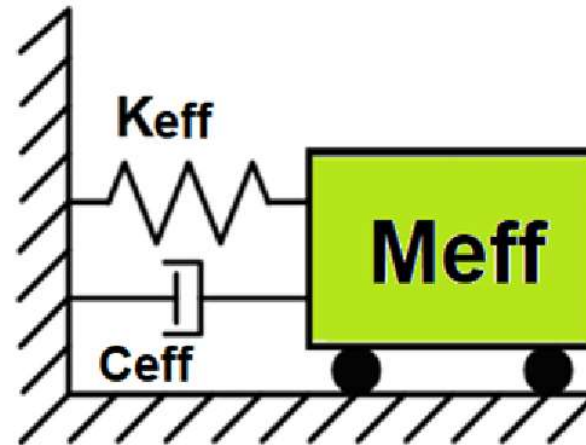


Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$\Delta_d$  = Lateral Displacement from Dynamic Analysis

$\Delta_y$  = Lateral Yield Displacement from Pushover Analysis

$\Delta_u$  = Lateral Ultimate Displacement from Pushover Analysis

12345678910111213141516171819202122232425262728293031323334

```
#####
#
#           >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#           NONLINEAR DYNAMIC ANALYSIS UNDER FREE-VIBRATION AND VISUALIZATION
#           RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING
#           INVESTIGATION OF THE EFFECTS OF MDOF STRUCTURAL DUCTILITY ON DAMAGE LEVEL AND STRUCTURAL
#           USING NONLINEAR DYNAMIC ANALYSIS
#
#           CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM
#
#IT IS RECOMMENDED FOR BETTER SEEING THE STRUCTURAL DUCTILITY AND STRUCTURAL OVER-STRENGTH
# INITIAL DISPLACEMENT HAVE TO BE GREATER THAN YIELD DISPLACEMENT.
#
#           THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#           EMAIL: salar.d.ghashghaei@gmail.com
#####
'''
Investigation of the Effects of Structural Ductility on Damage Level and Structural Damping
This code implements a comprehensive nonlinear dynamic analysis framework for
performance-based earthquake engineering assessment of single-degree-of-freedom
(SDOF) systems. The methodology combines traditional nonlinear time-history
analysis with modern probabilistic and machine learning techniques for advanced
structural performance evaluation with changing structural ductility ratio and over strength

KEY ENGINEERING OBJECTIVES:
1. Comparative assessment of hysteretic models for free-vibration response prediction
2. Probabilistic free-vibration demand analysis using multiple ground motions
3. Development of fragility curves for performance-based earthquake engineering
4. Integration of data science methods for structural reliability assessment

ANALYTICAL FEATURES:
- Nonlinear material behavior with pinching and degradation
- Response spectrum analysis across period range
'''
```

17%

Python Console

Files

Help

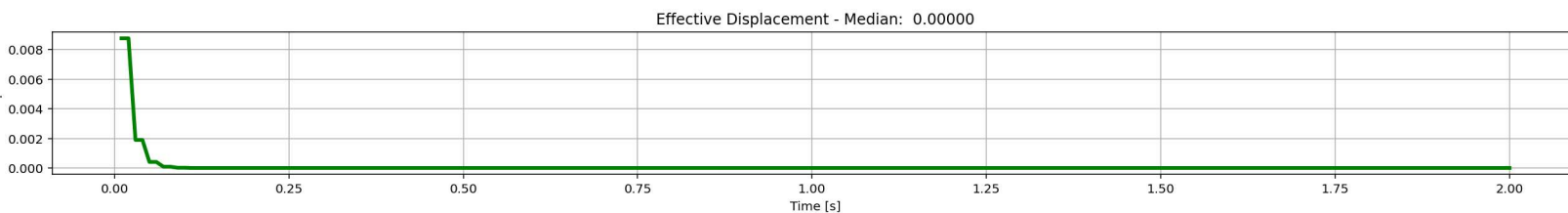
Variable Explorer

Debugger

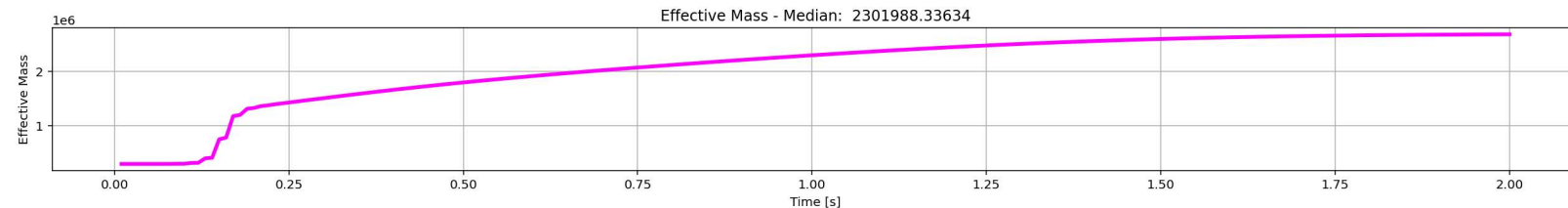
Plots

History

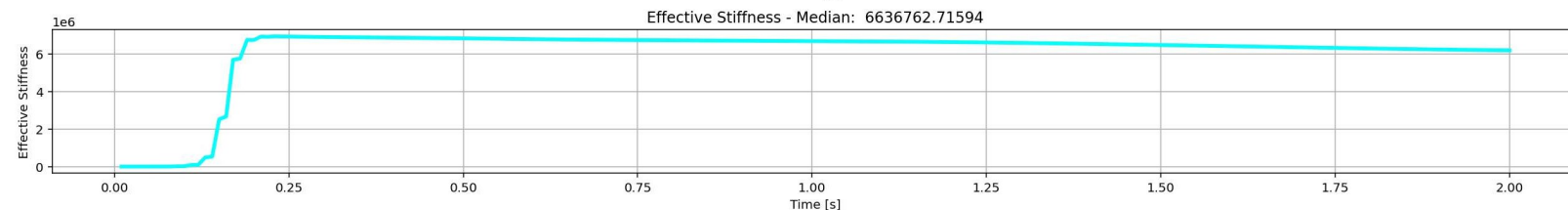
Line 248 Col 26 LITE-S CPU-RW Mem 60%



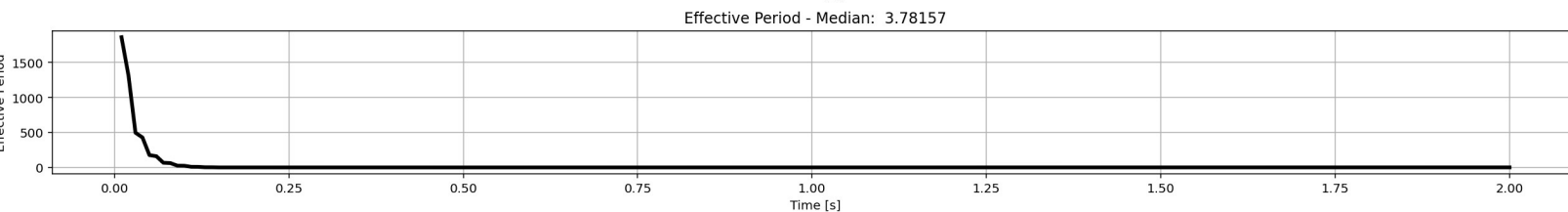
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



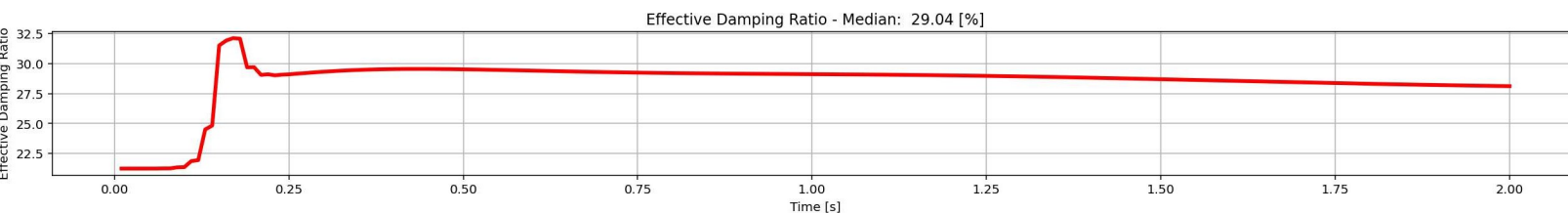
$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$



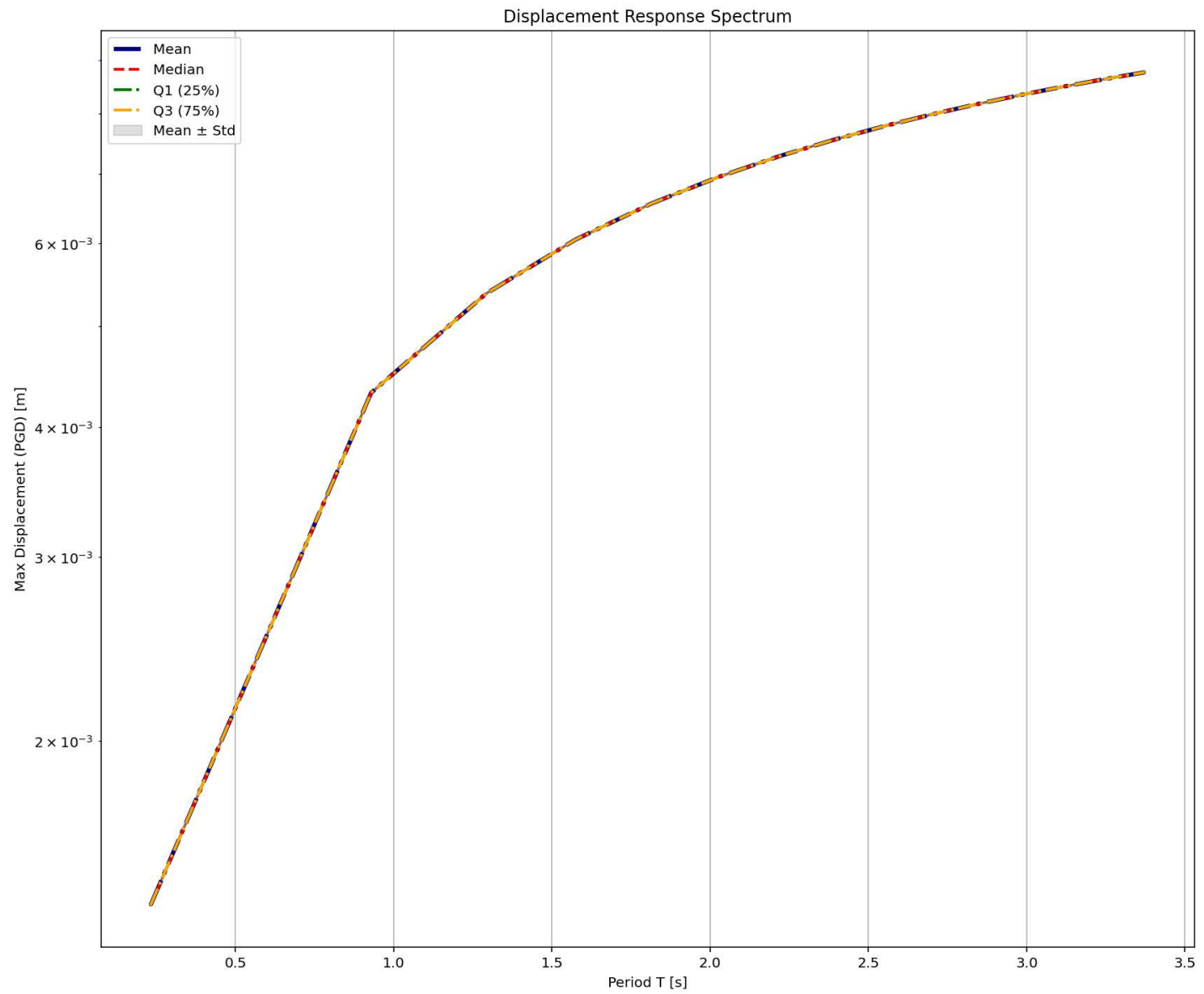
$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$



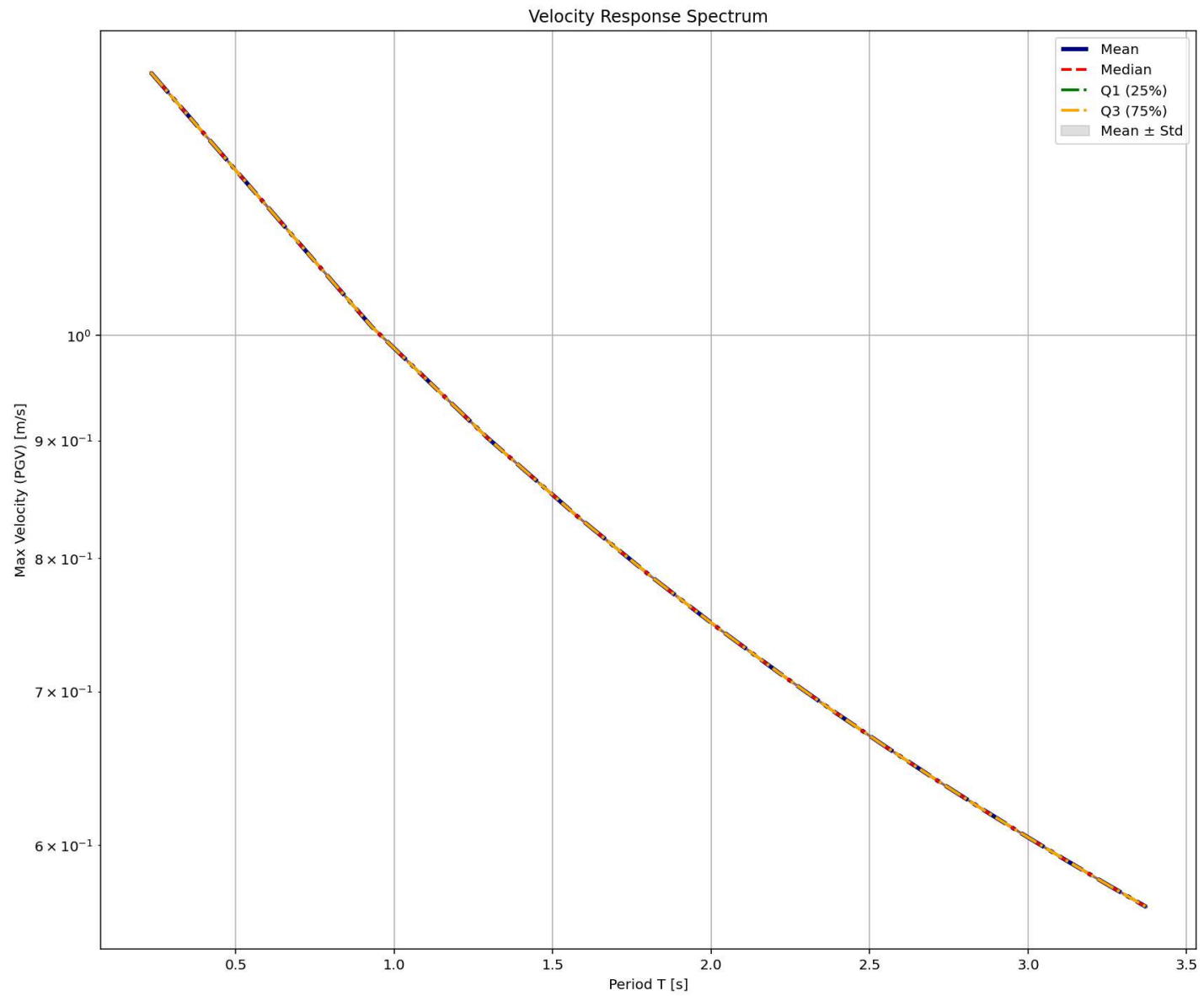
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



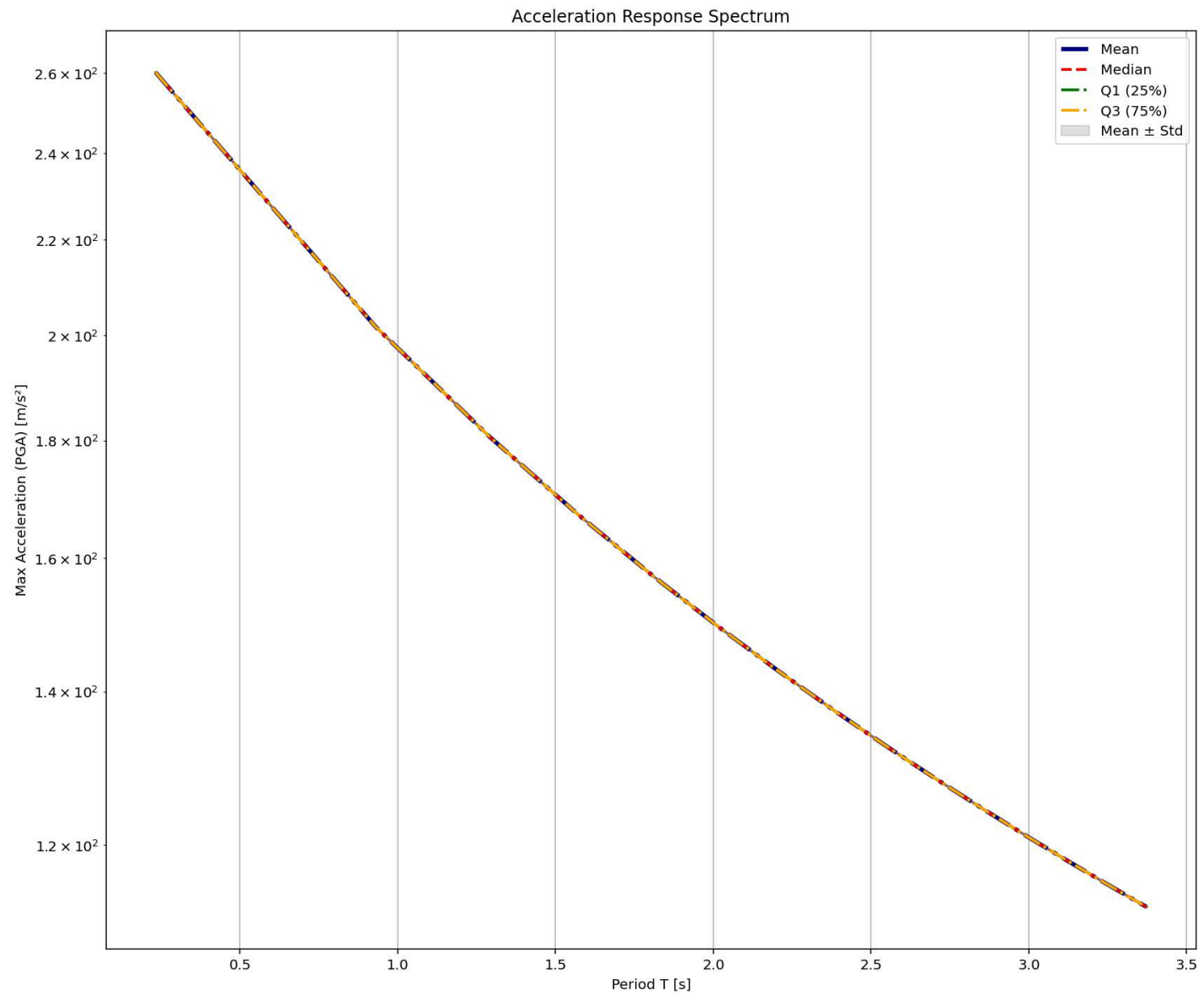
$$\xi_{eff} = \frac{\sum_{i=1}^n \xi_i \Delta_i}{\Delta_d}$$

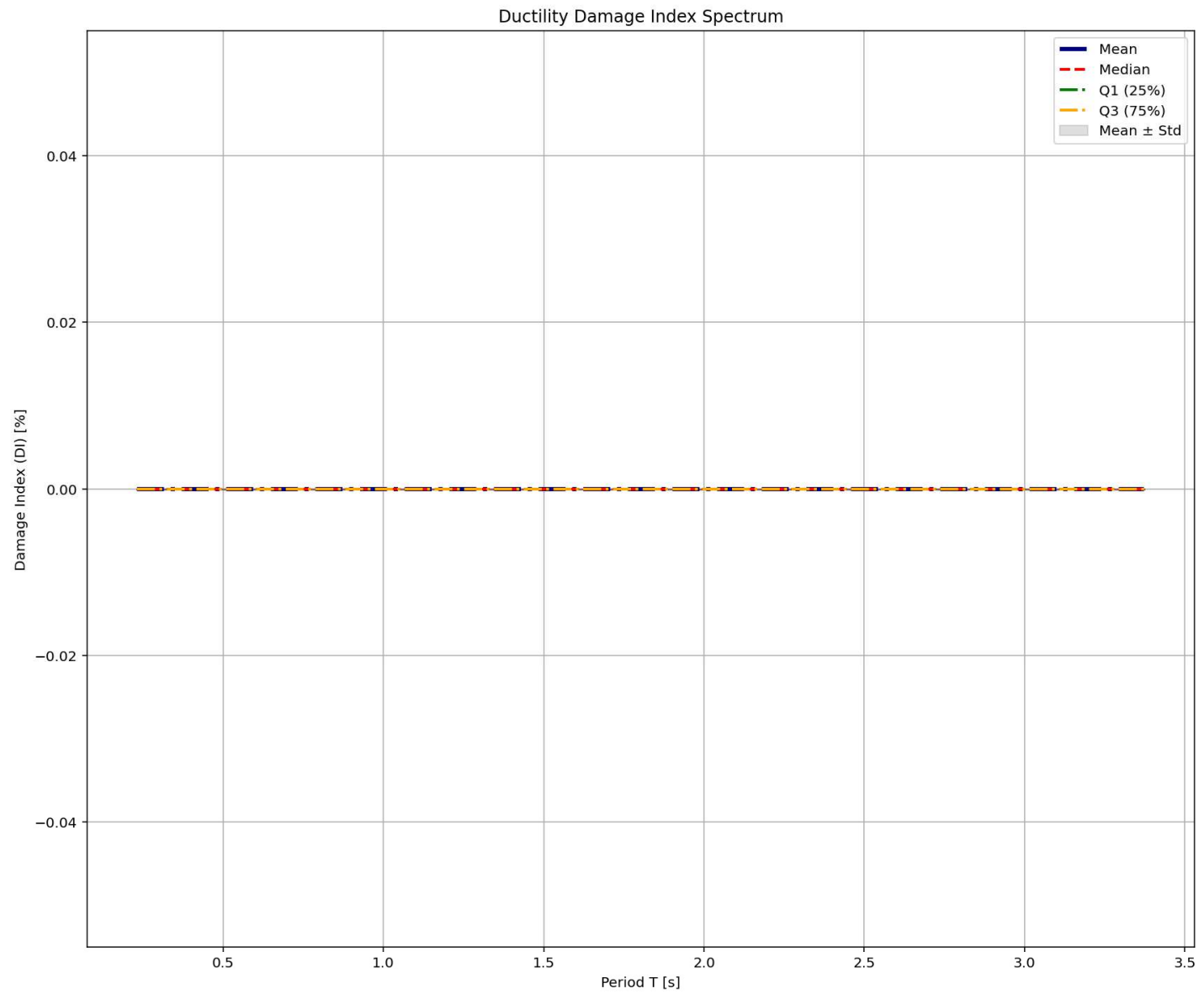


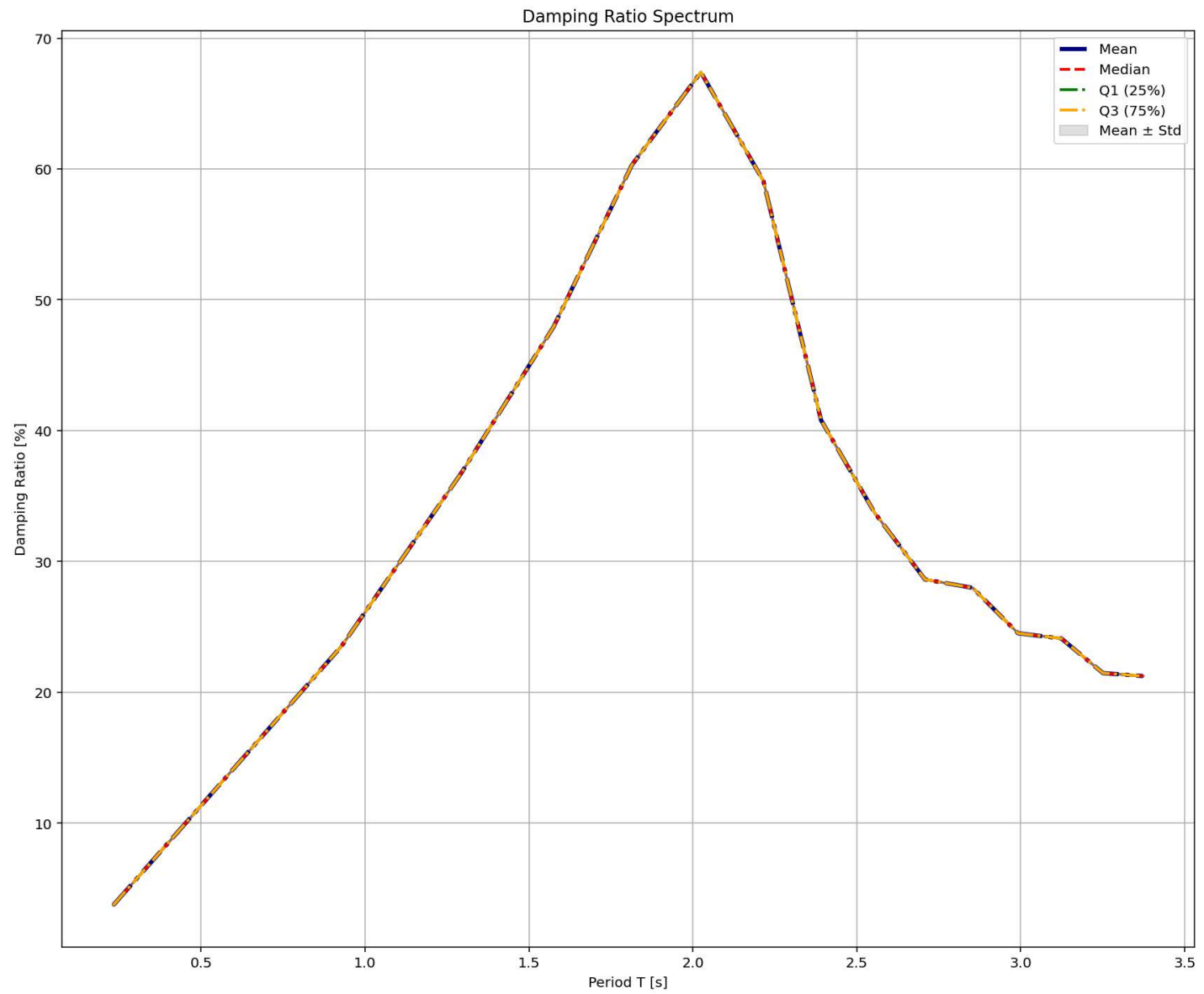


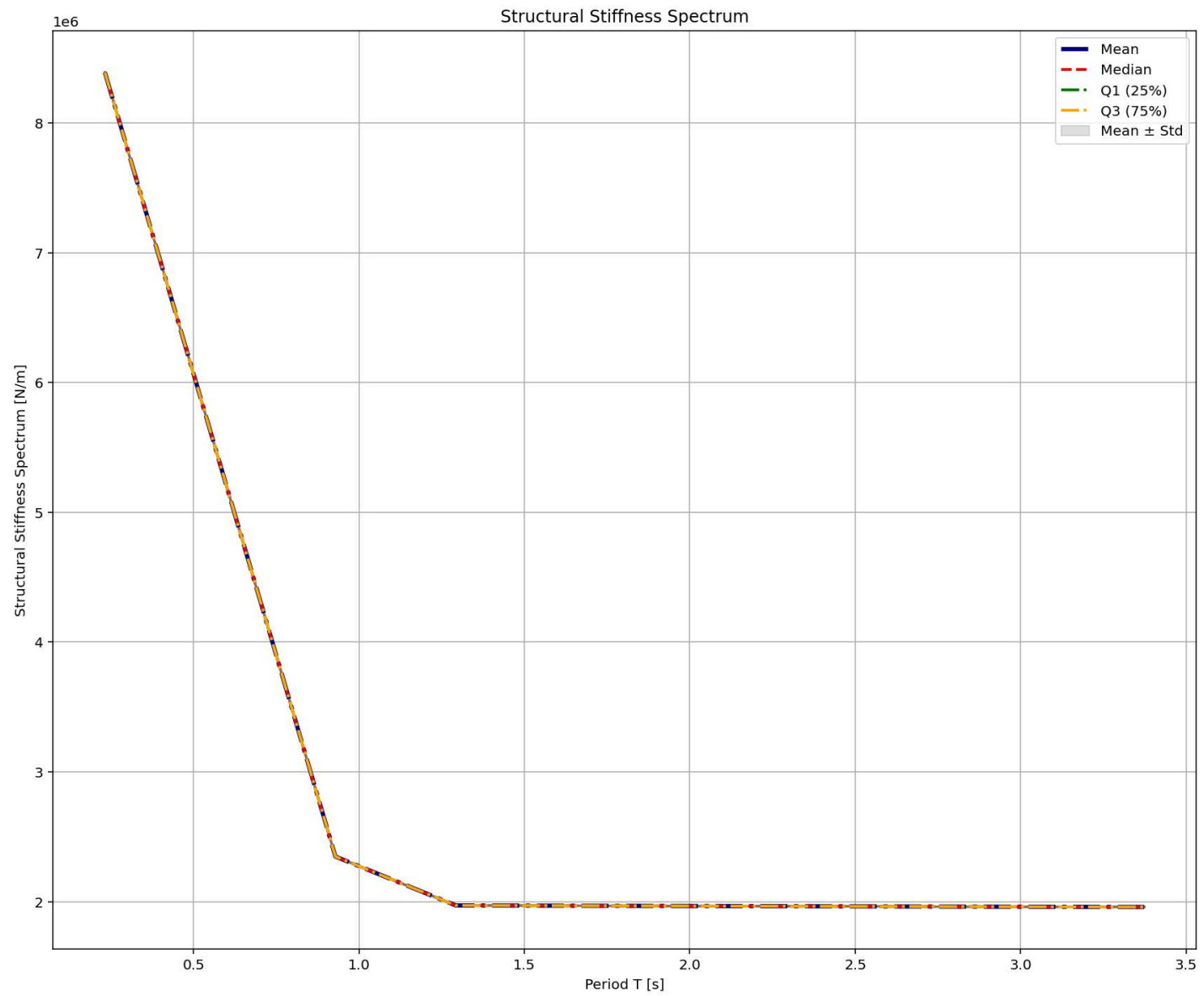


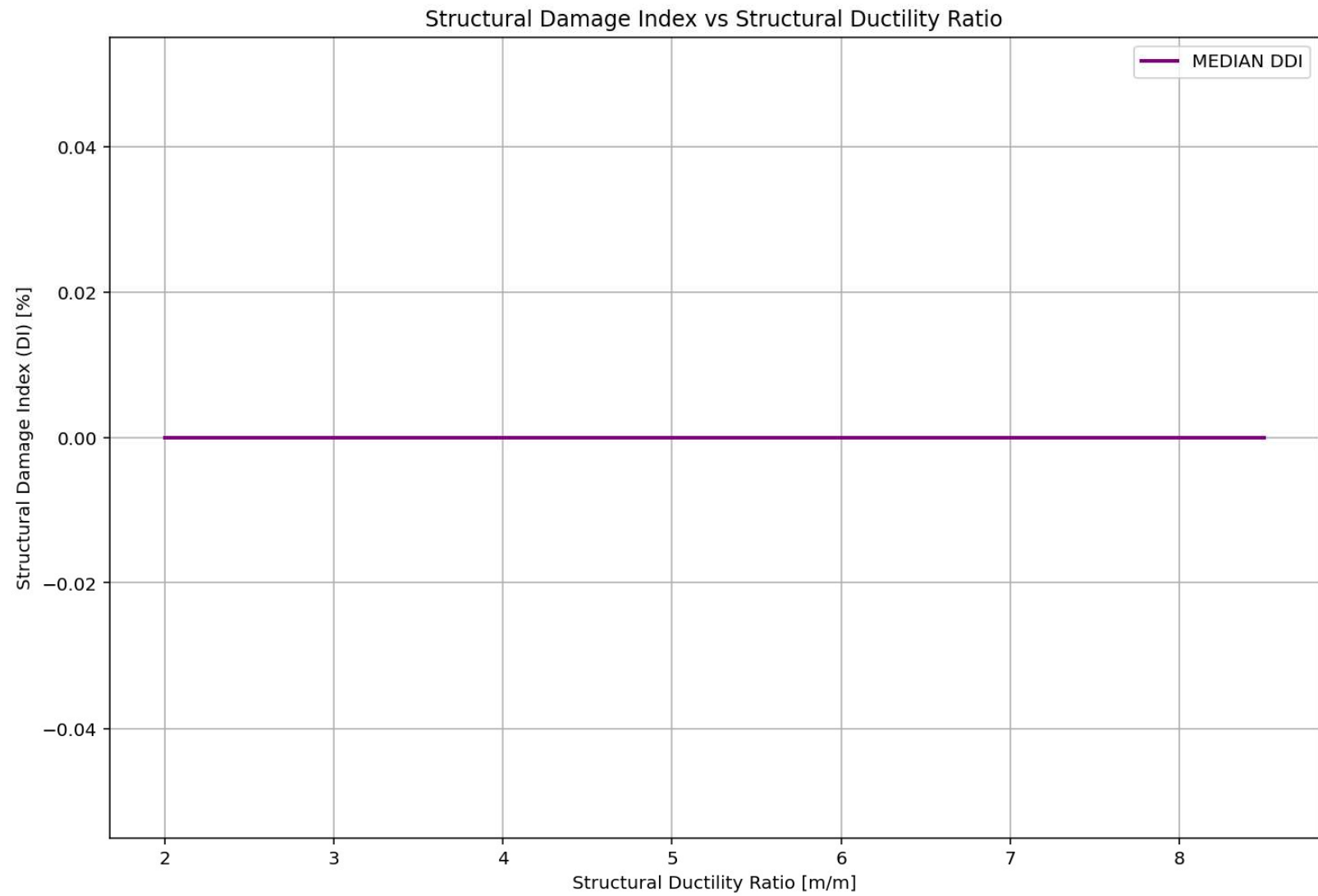


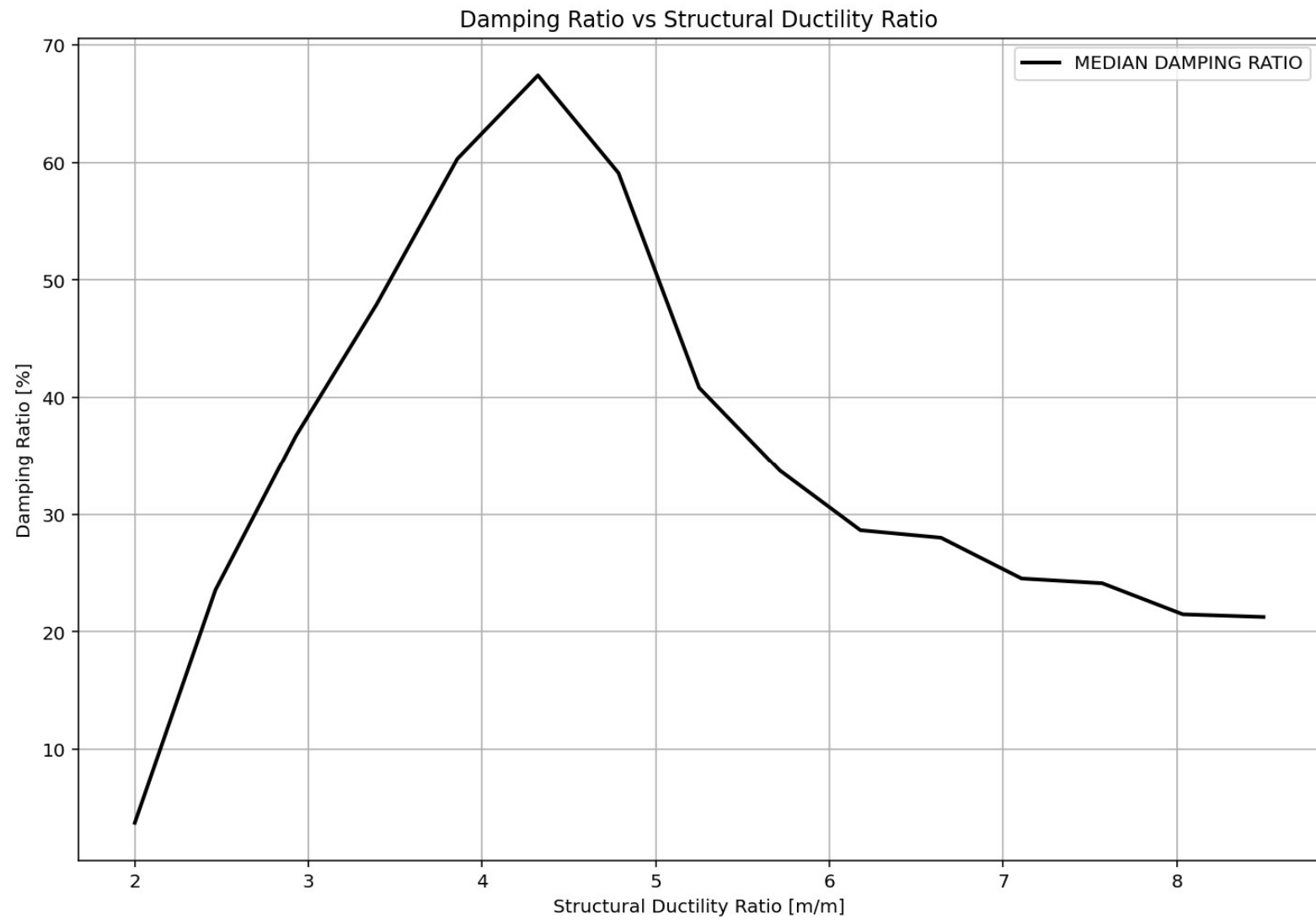




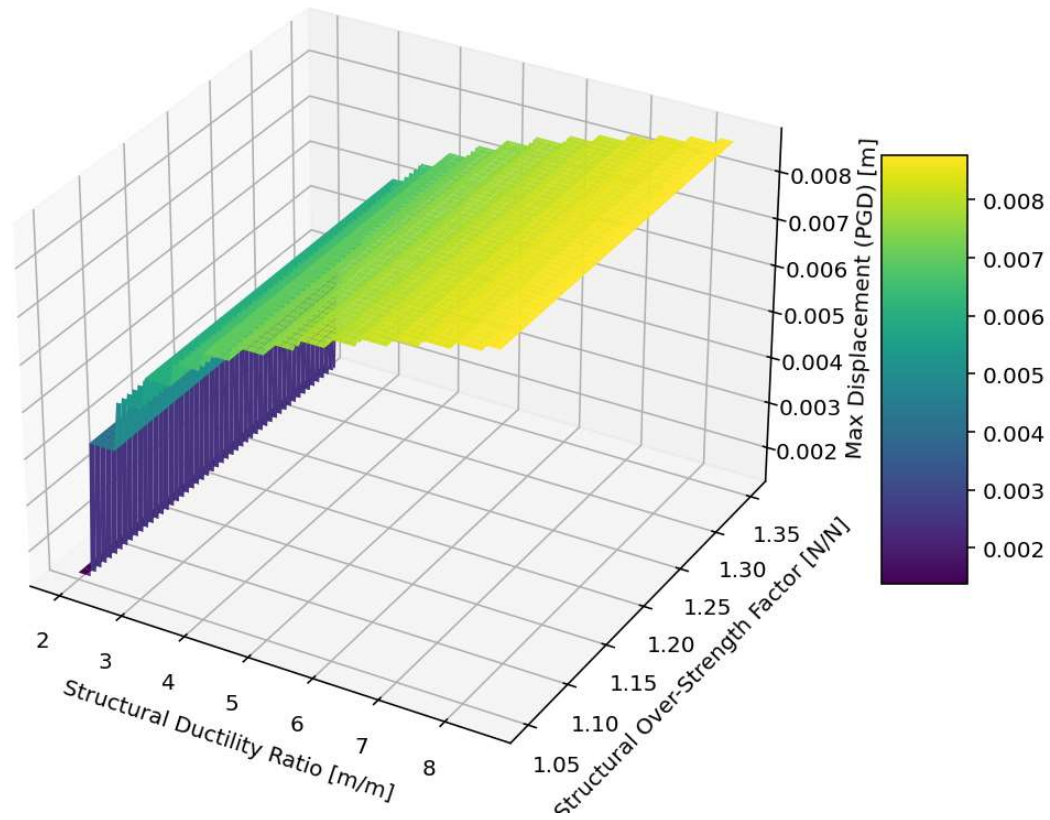






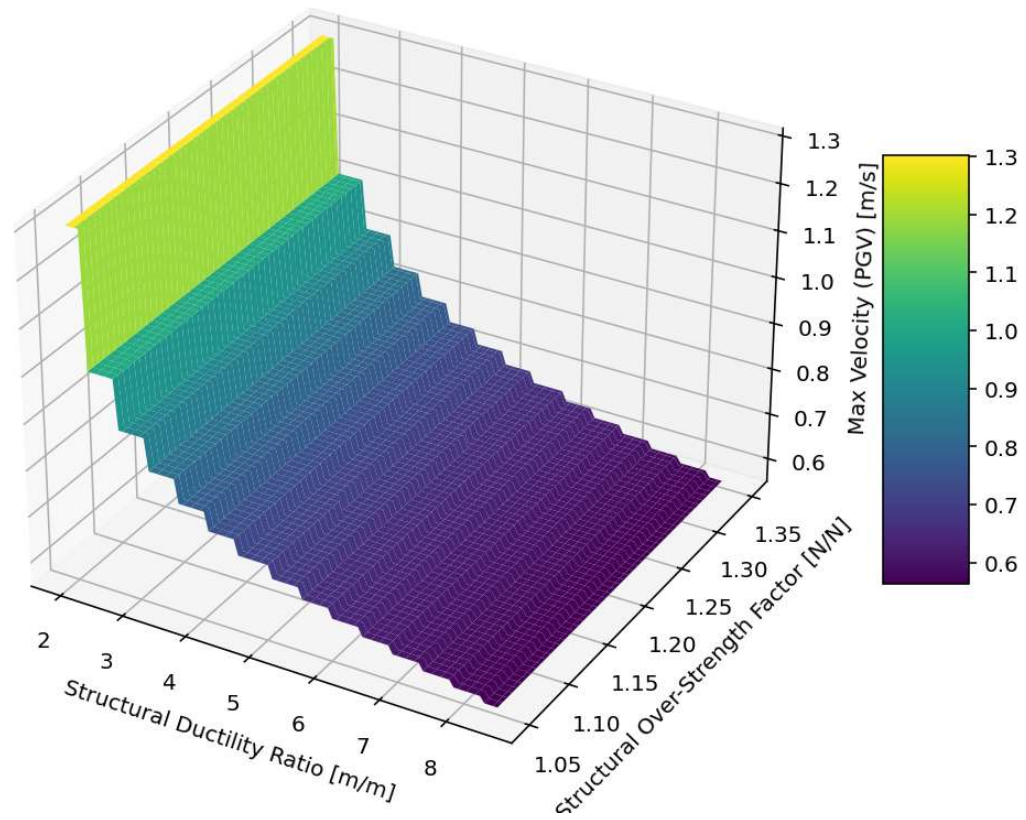


3D Contour Plot of Max Displacement (PGD) [m]

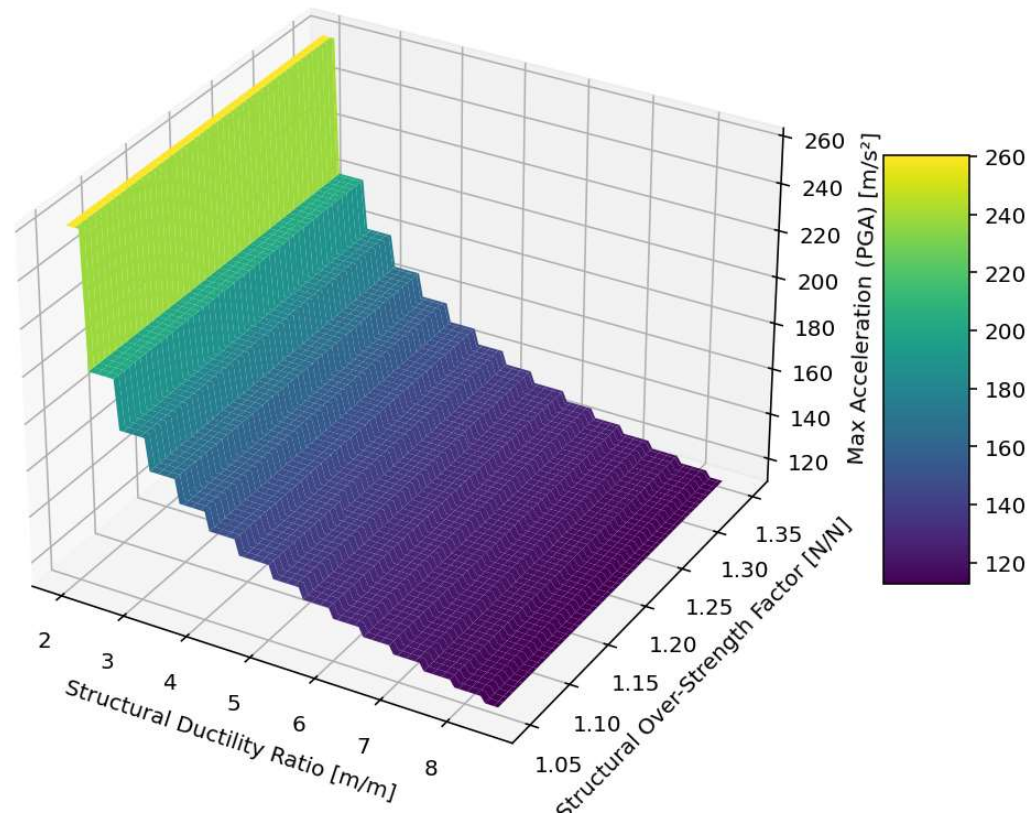




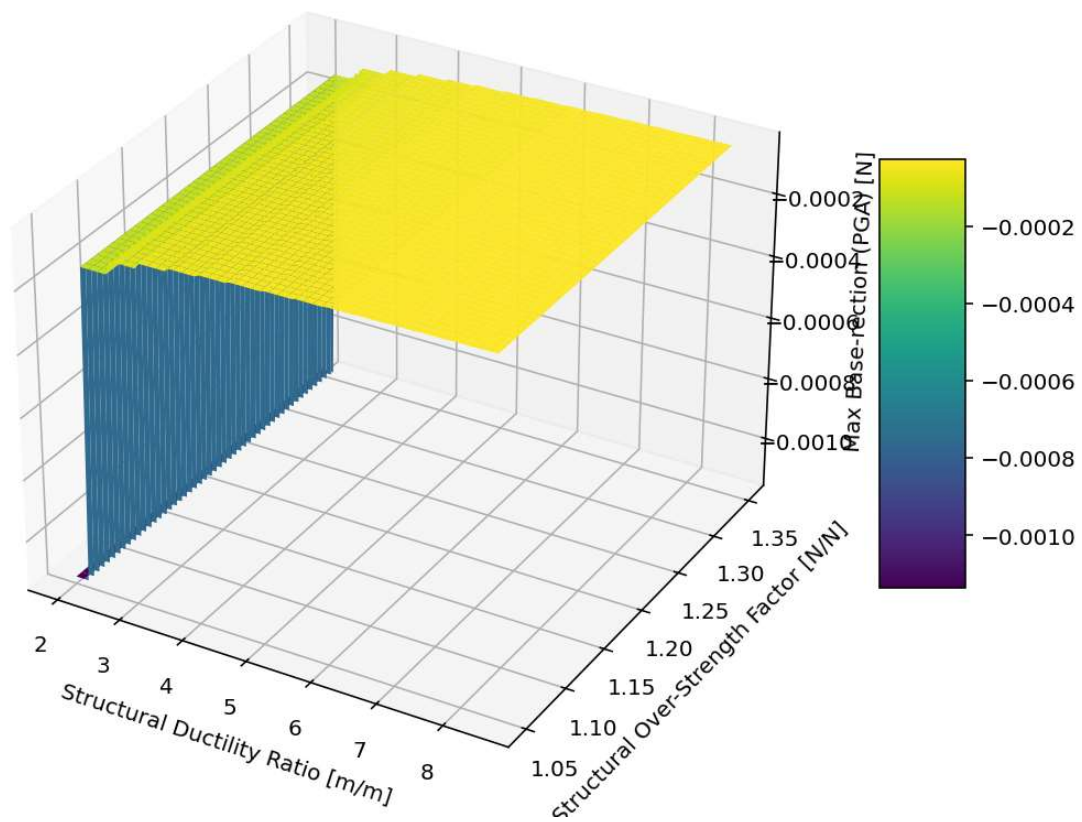
3D Contour Plot of Max Velocity (PGV) [m/s]



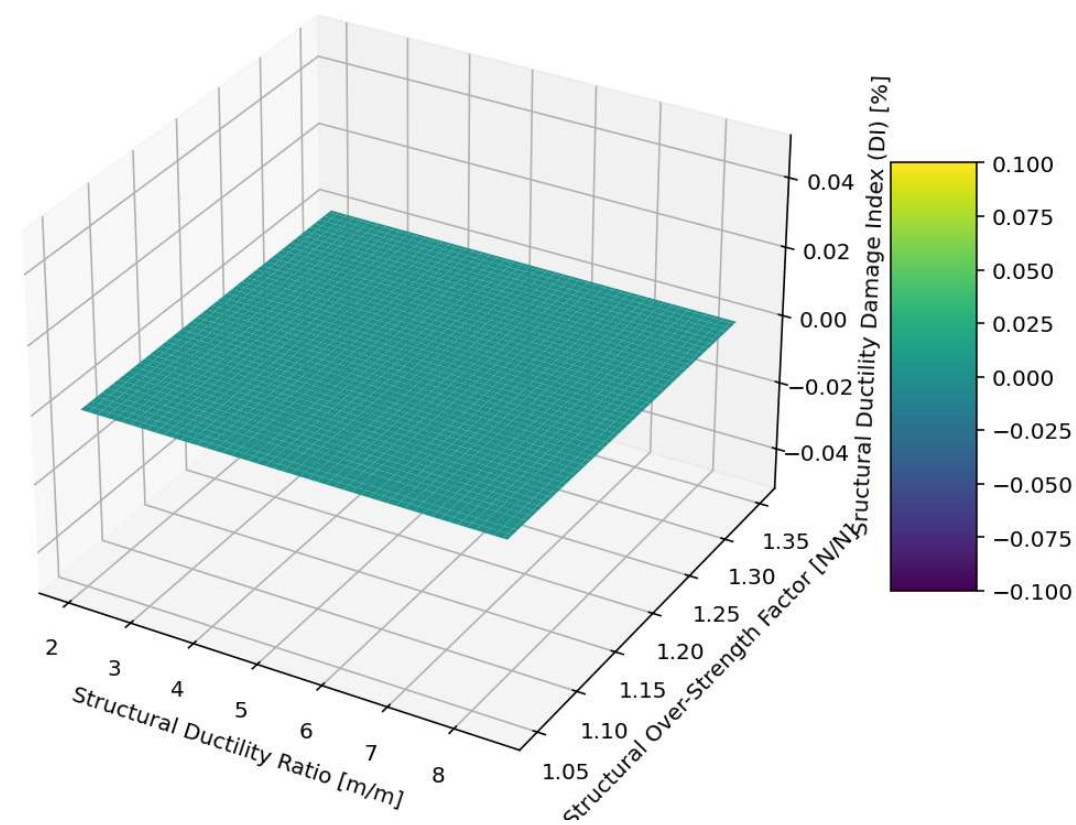
3D Contour Plot of Max Acceleration (PGA) [ $\text{m/s}^2$ ]



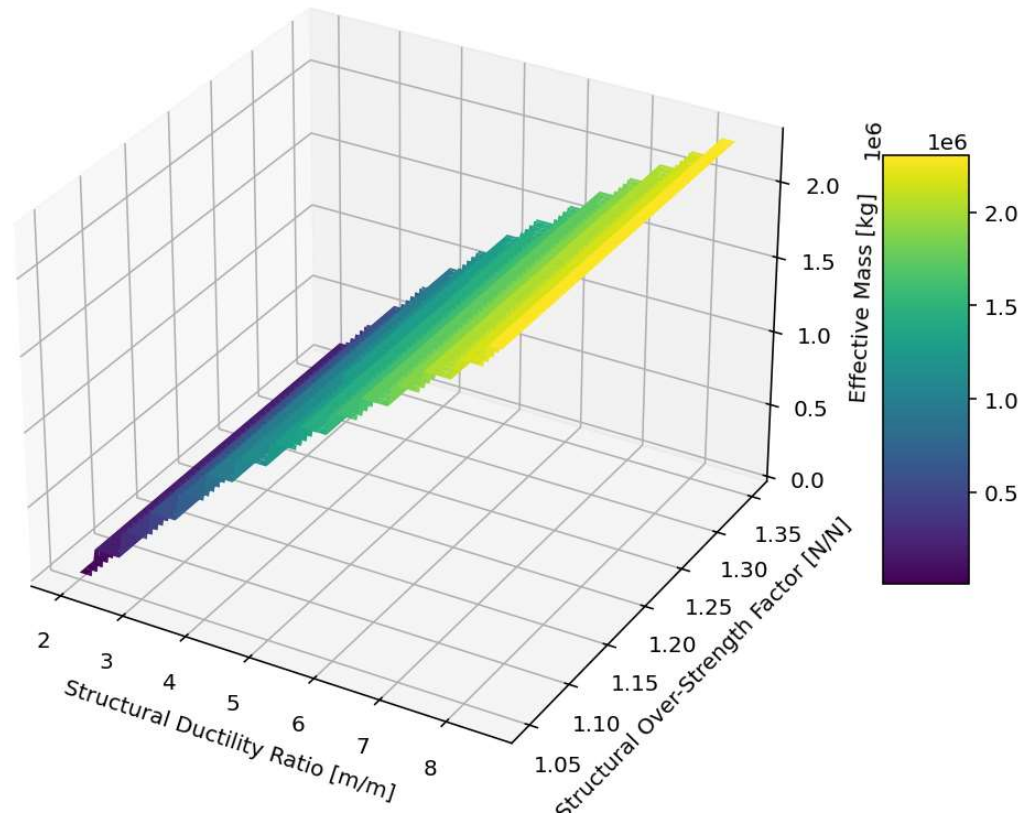
3D Contour Plot of Max Base-rection (PGA) [N]



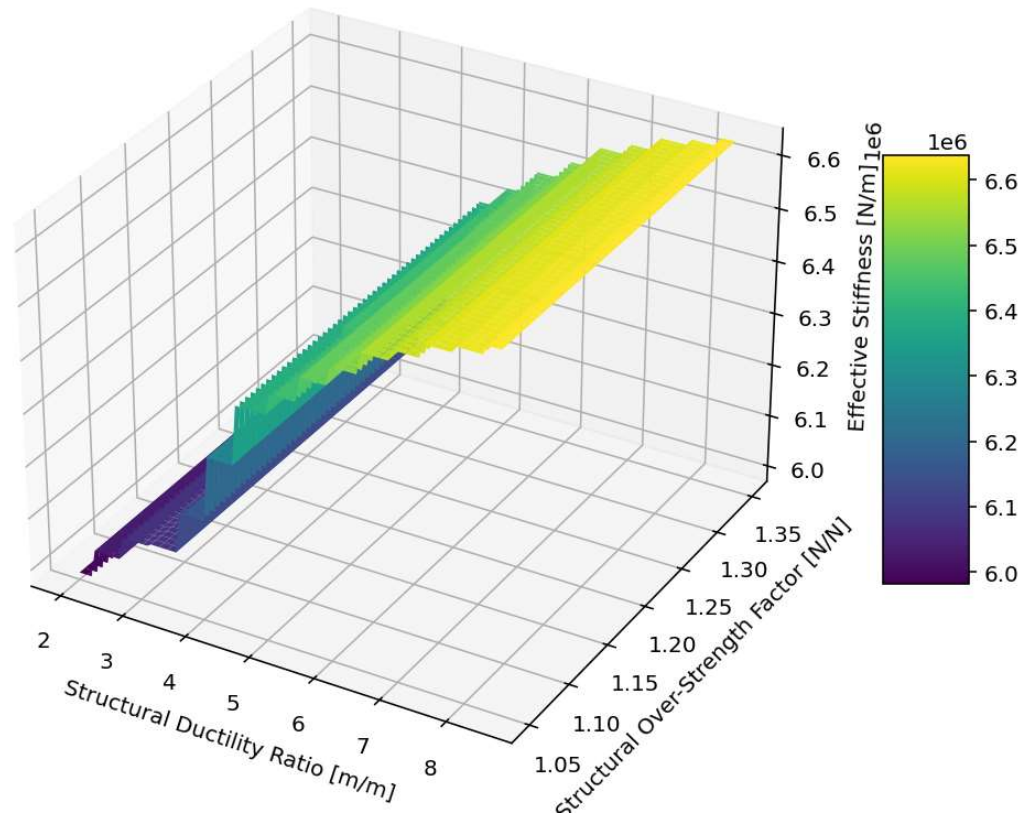
3D Contour Plot of Sructural Ductility Damage Index (DI) [%]



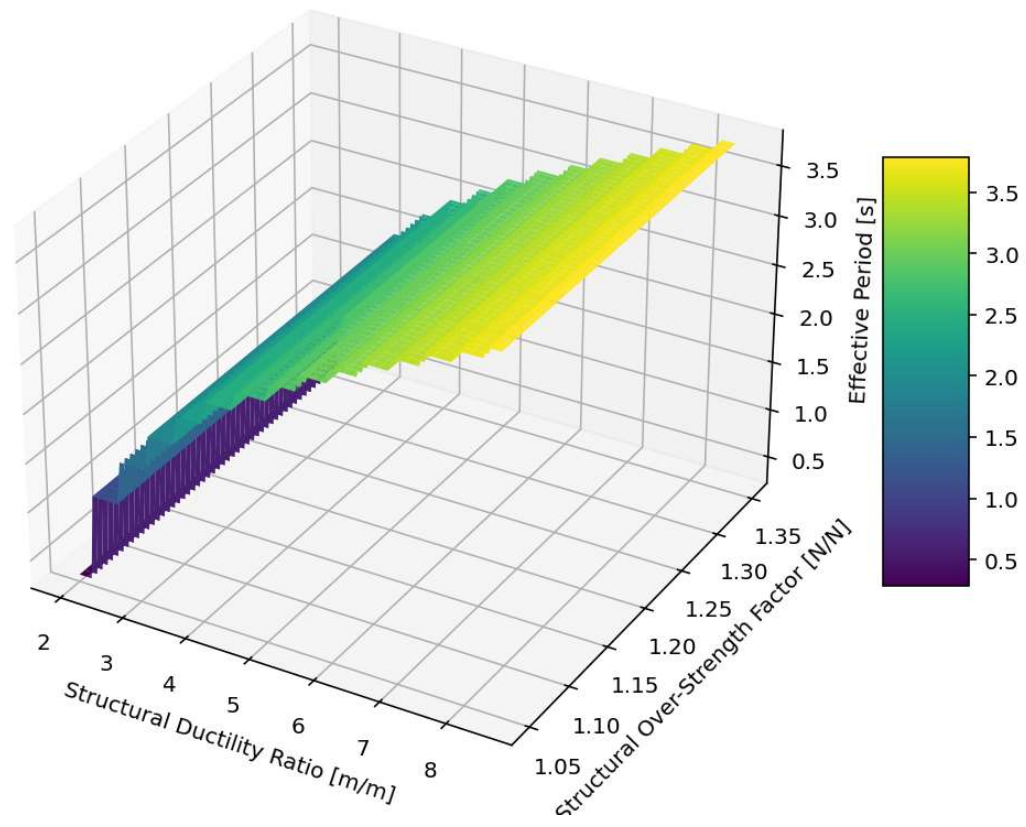
3D Contour Plot of Effective Mass [kg]



3D Contour Plot of Effective Stiffness [N/m]

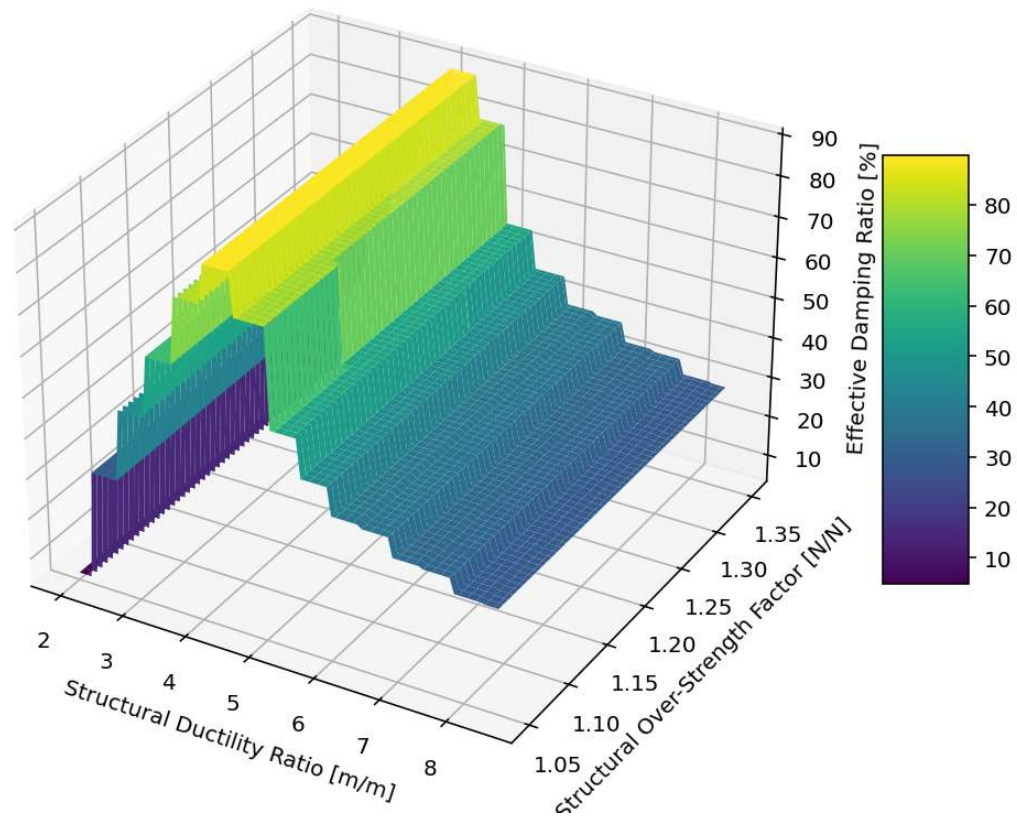


3D Contour Plot of Effective Period [s]

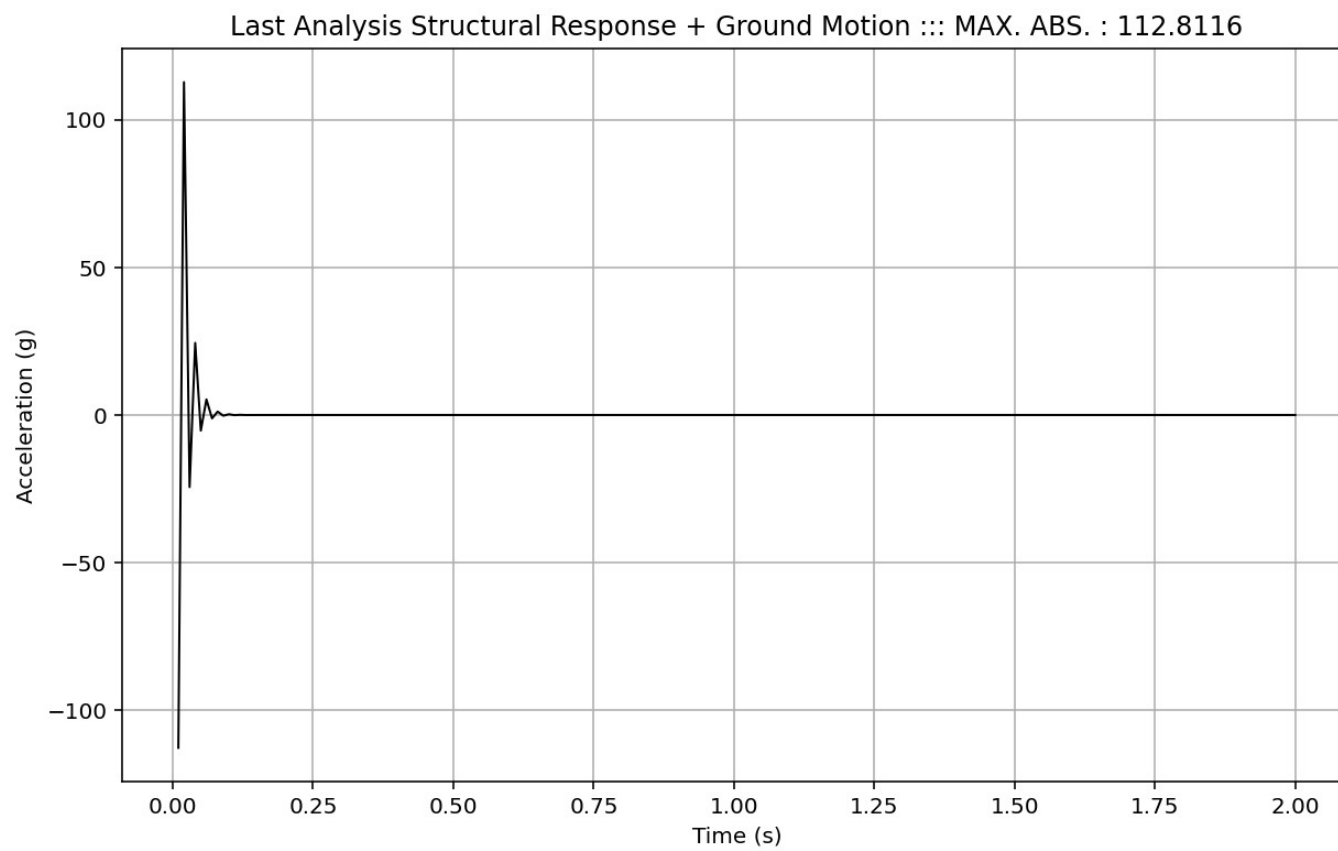


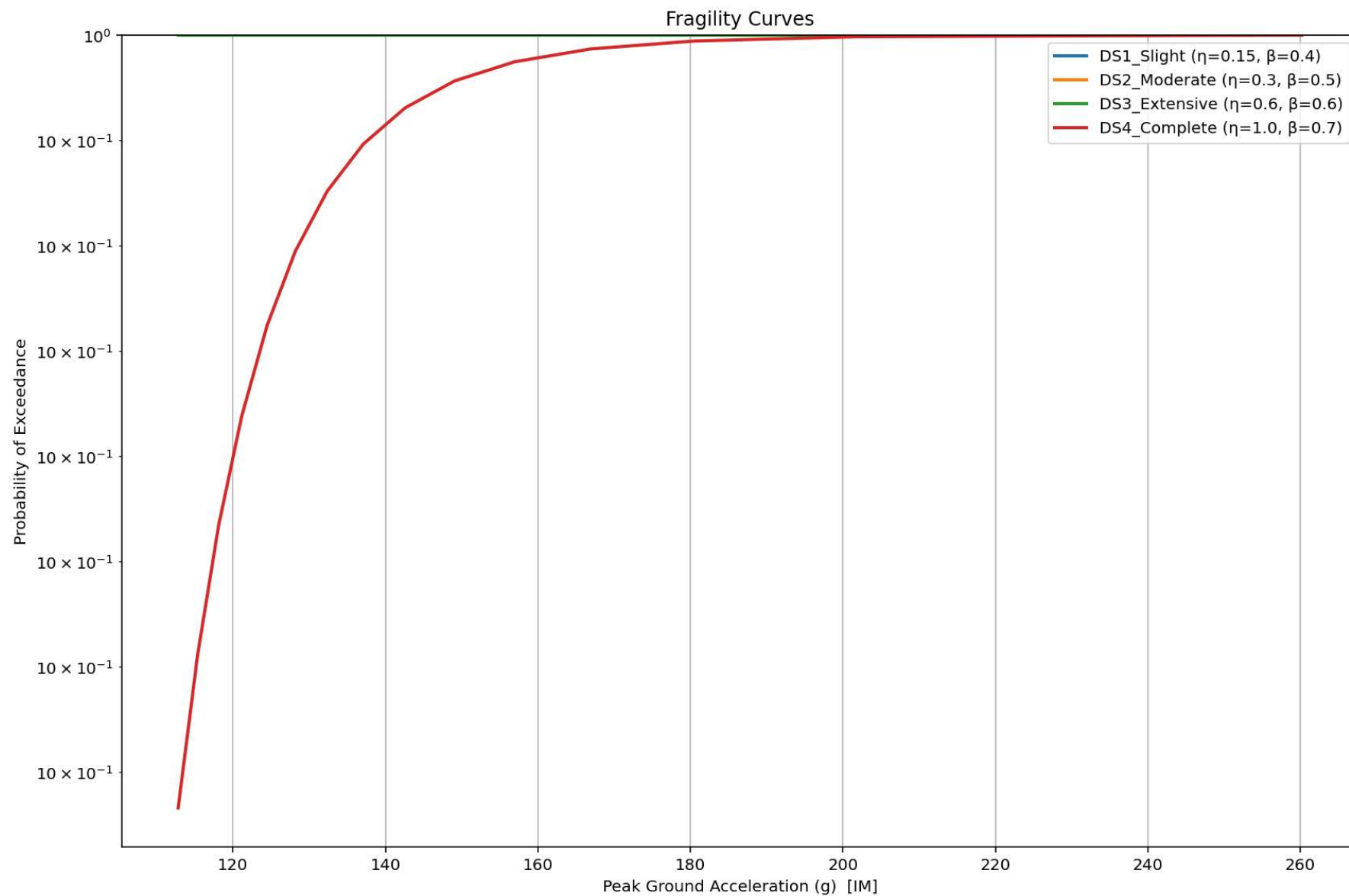


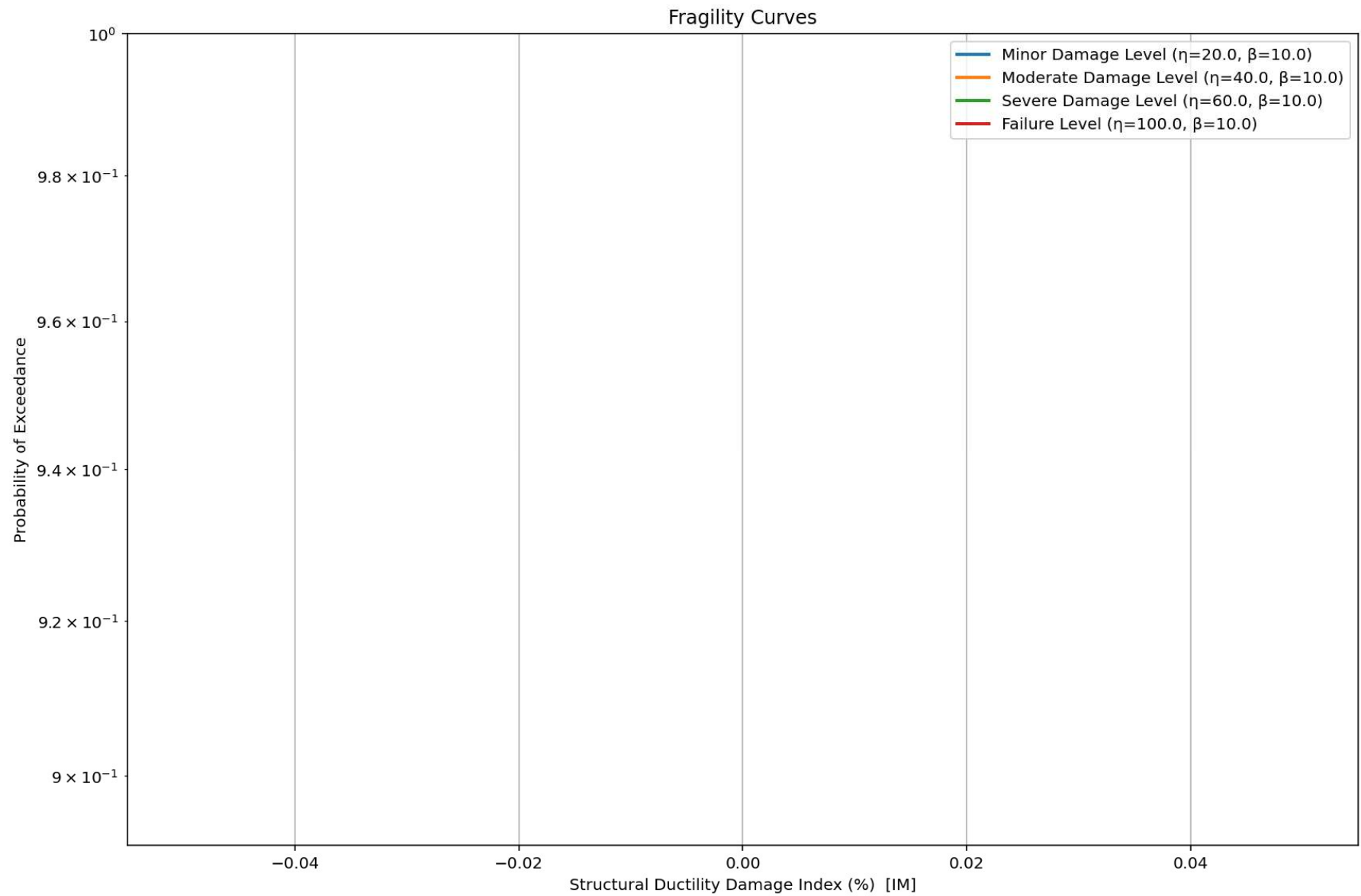
3D Contour Plot of Effective Damping Ratio [%]



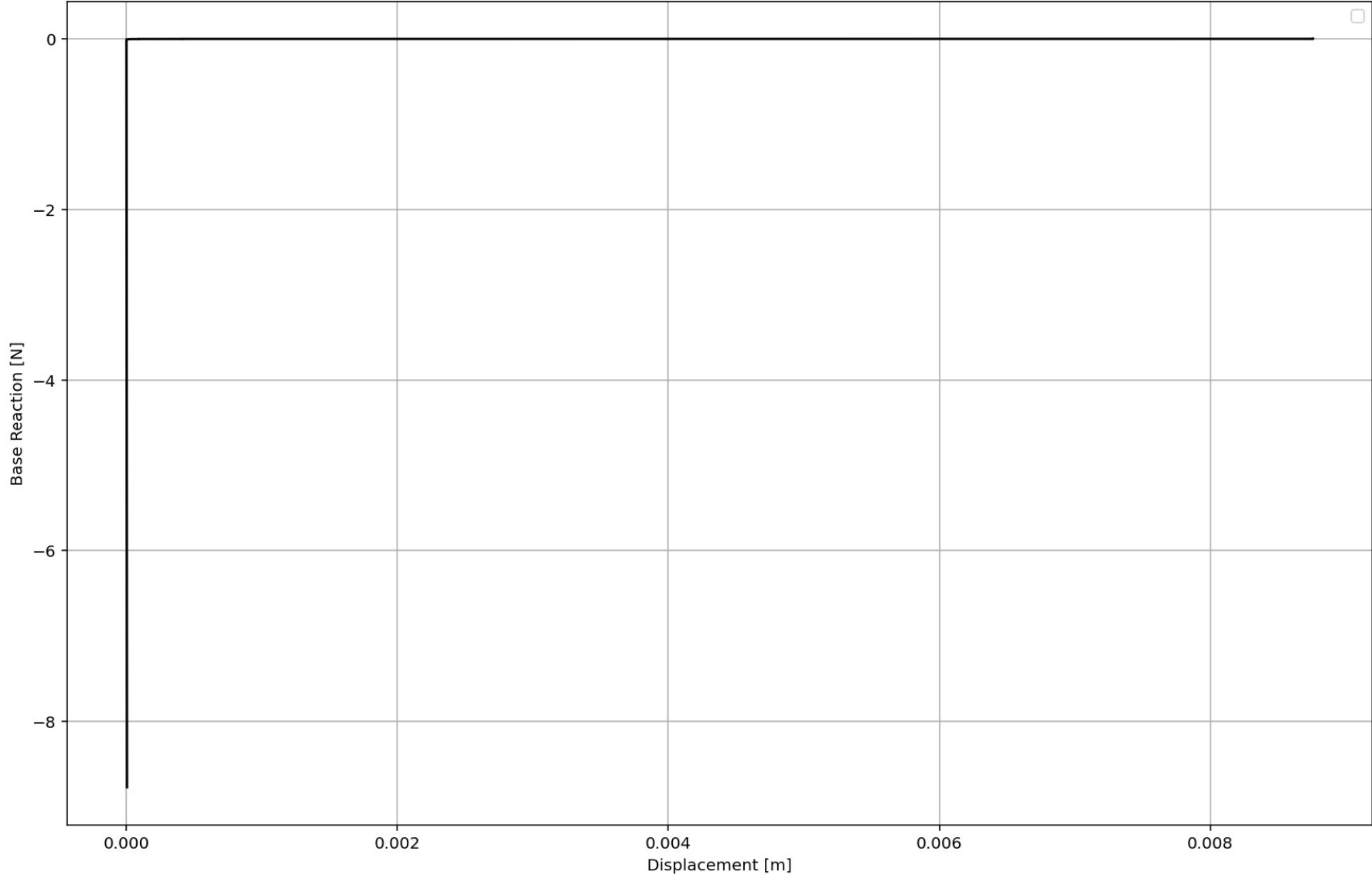




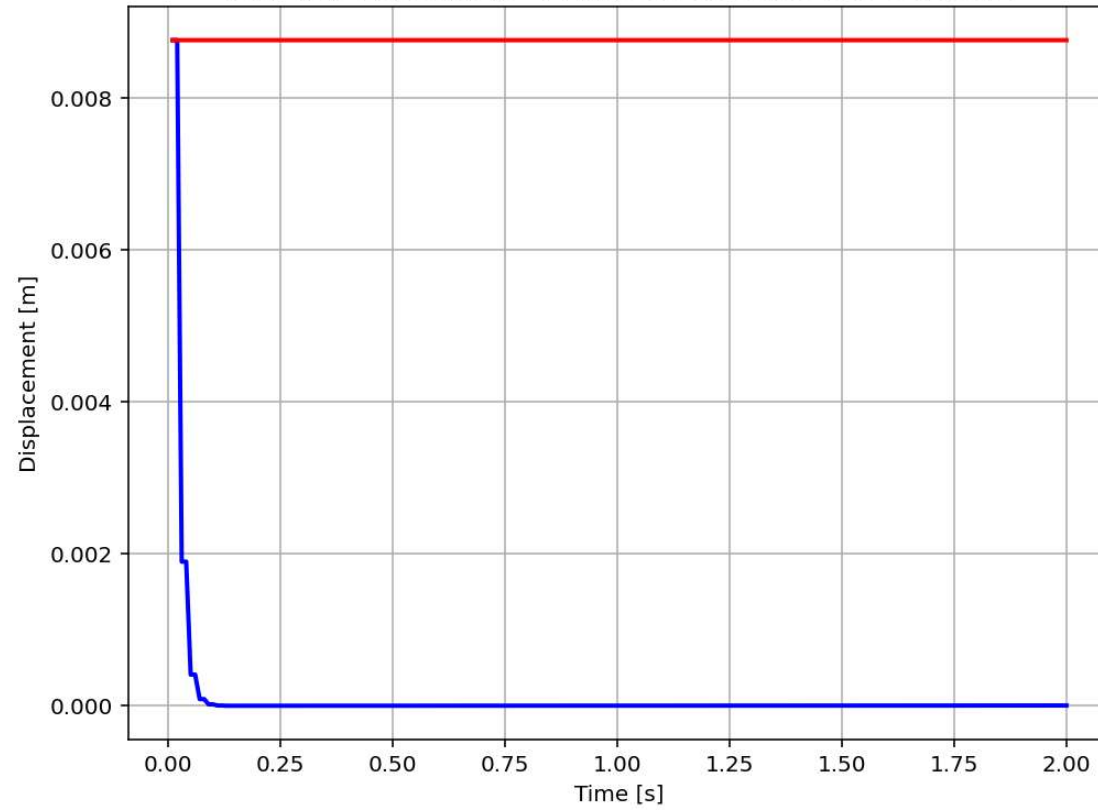


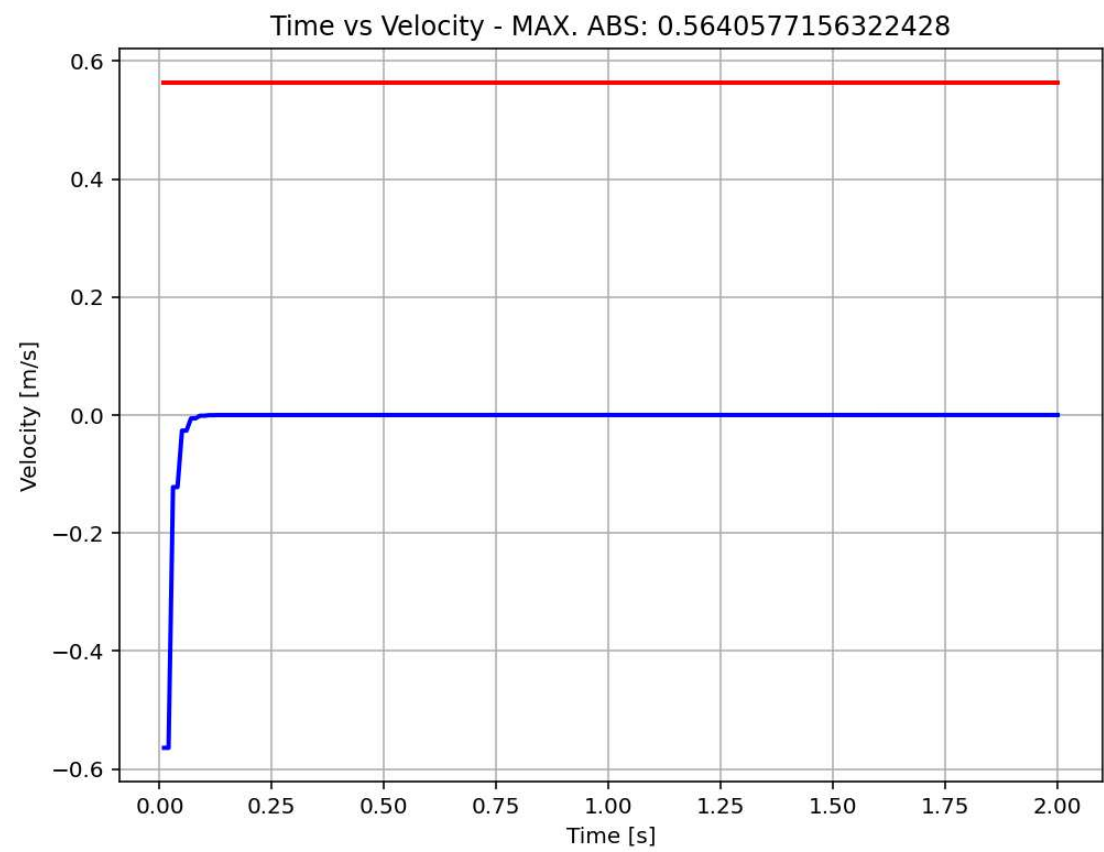


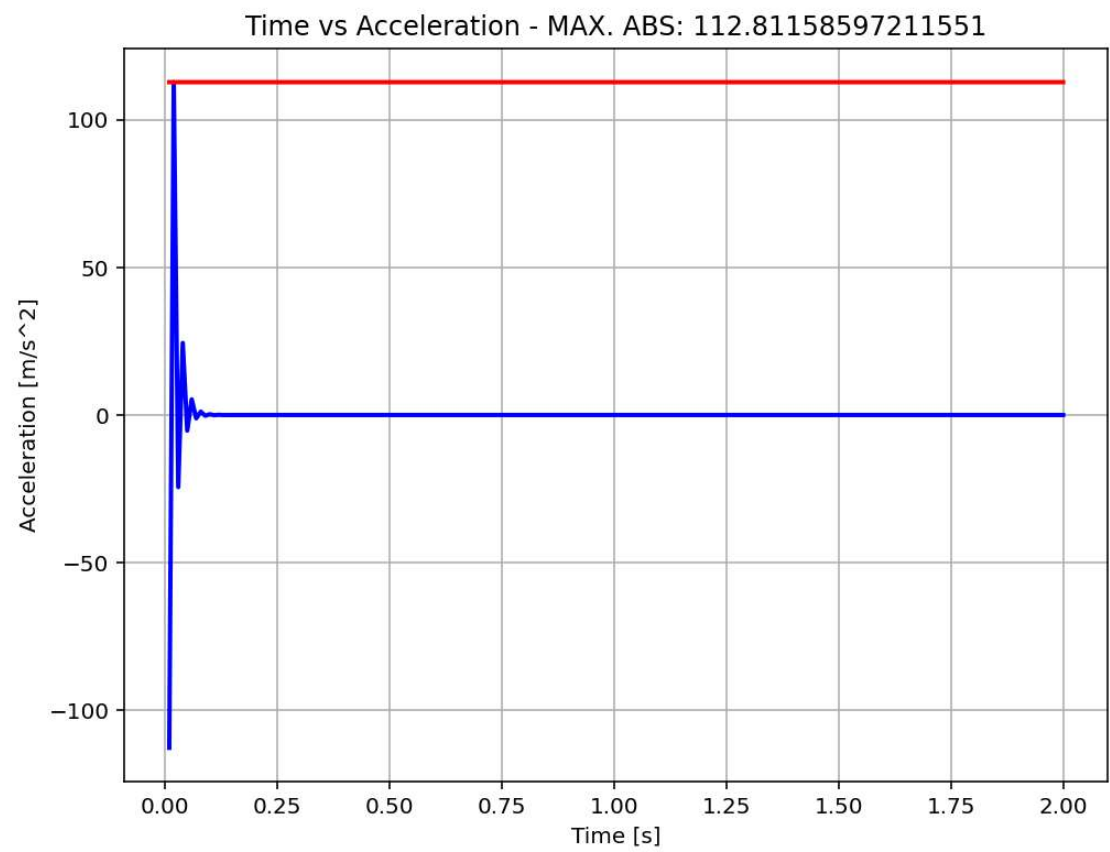
Displacement & Base Reaction Relation From Last Dynamic Analysis

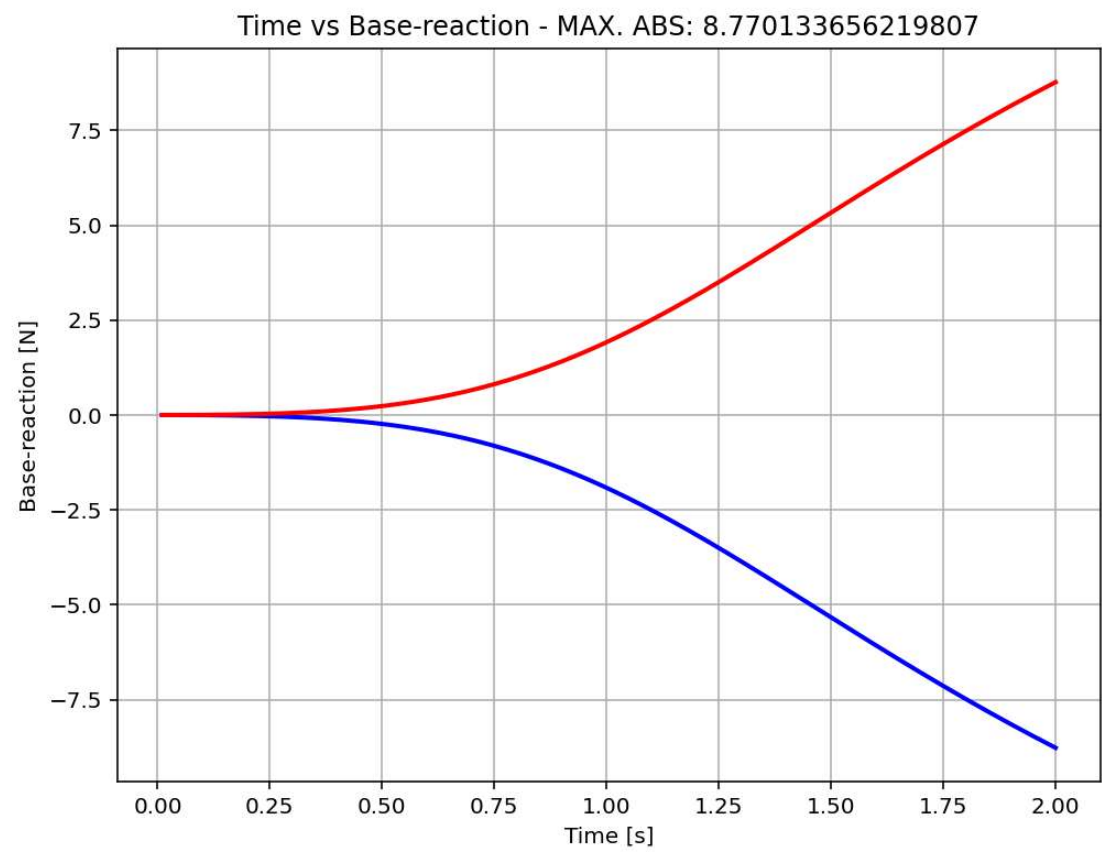


Time vs Displacement - MAX. ABS: 0.008759424985960919

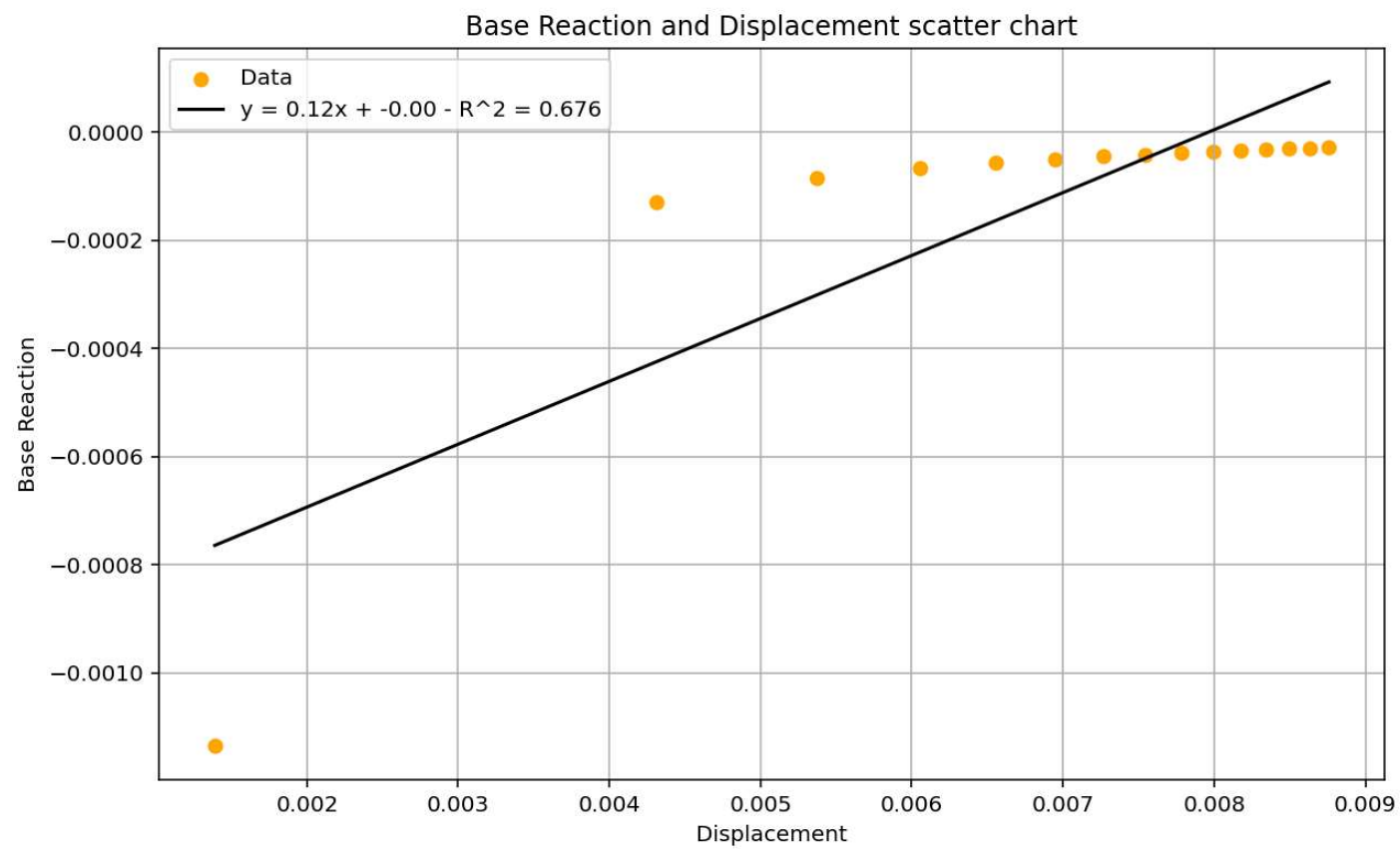


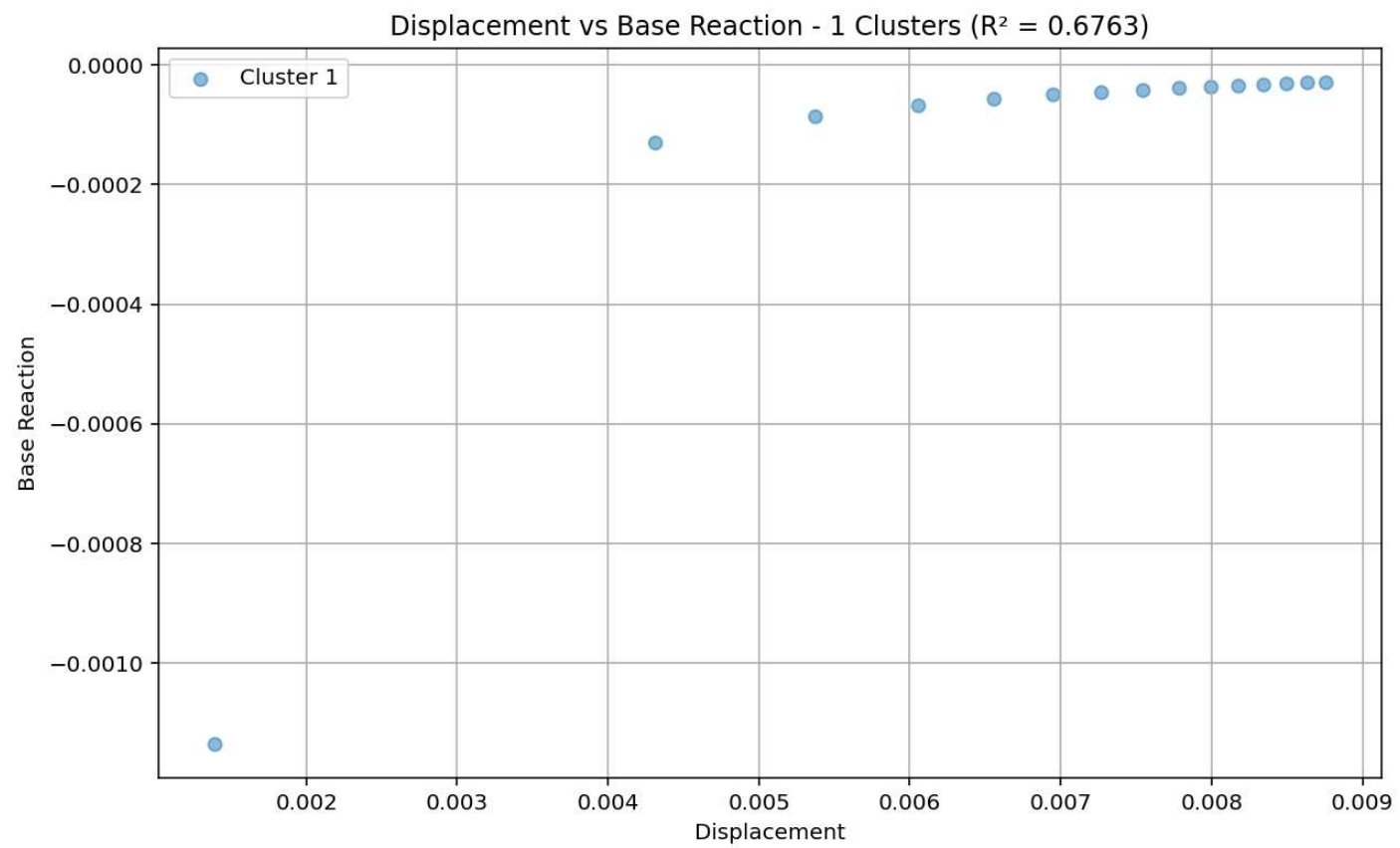


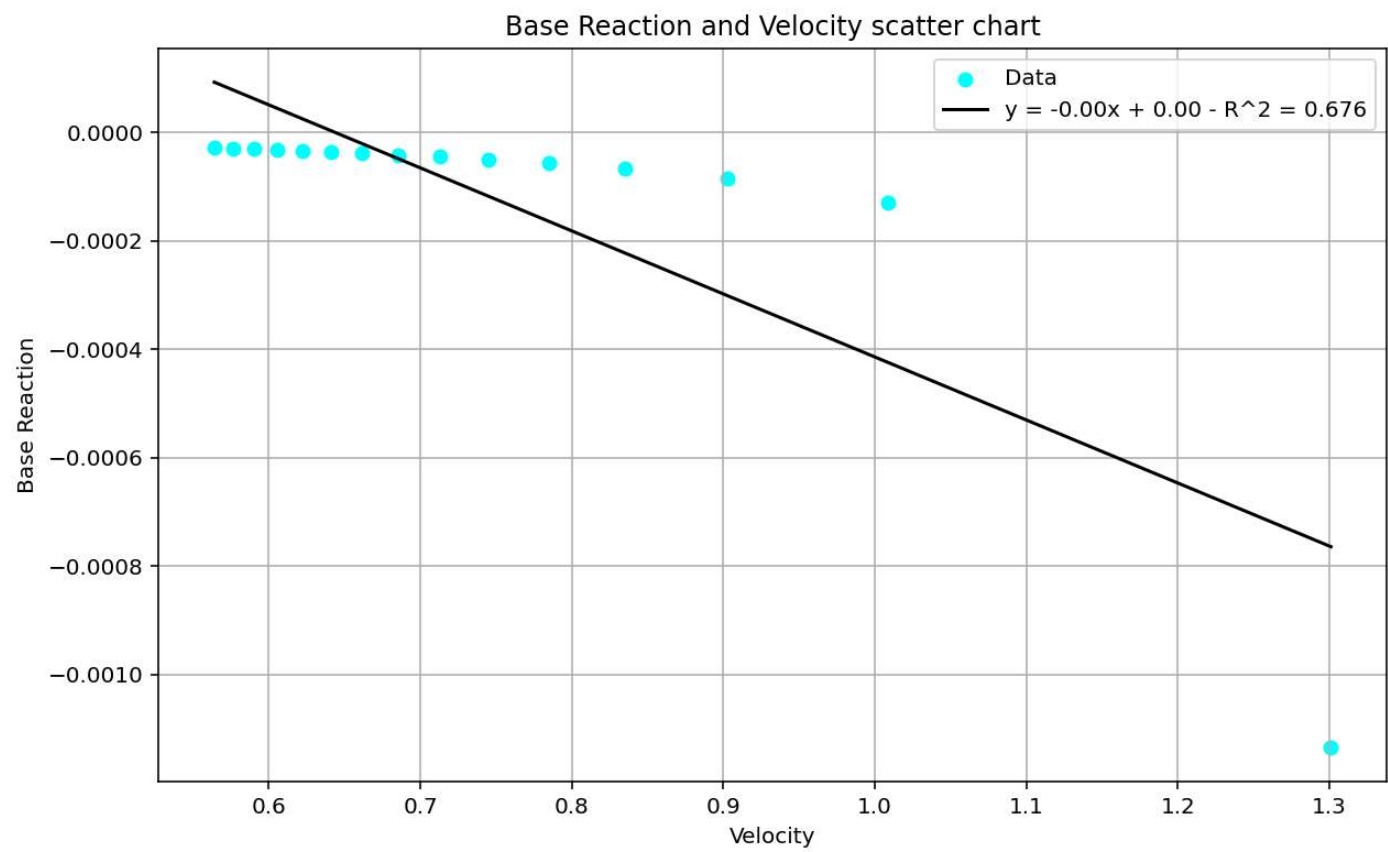


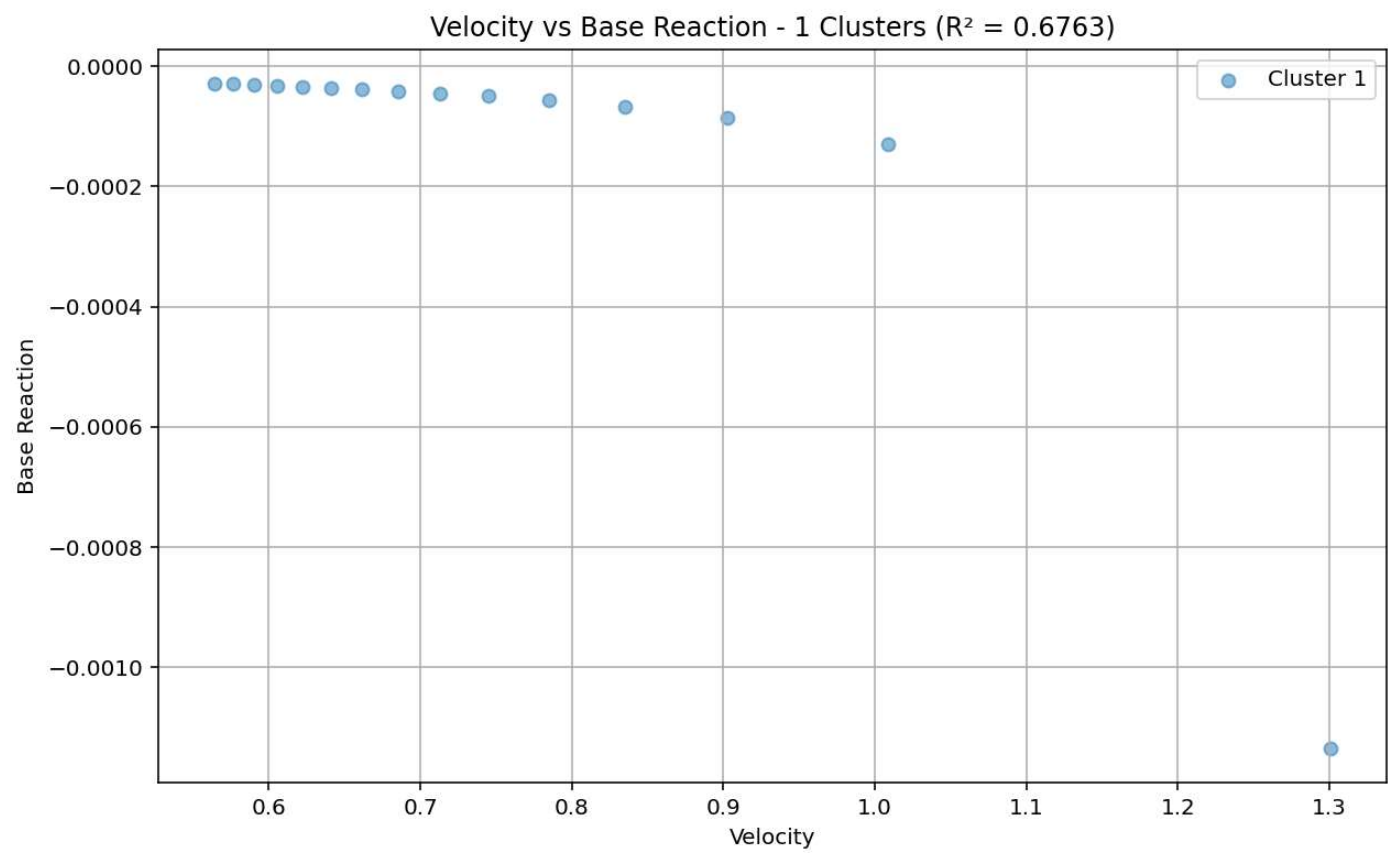


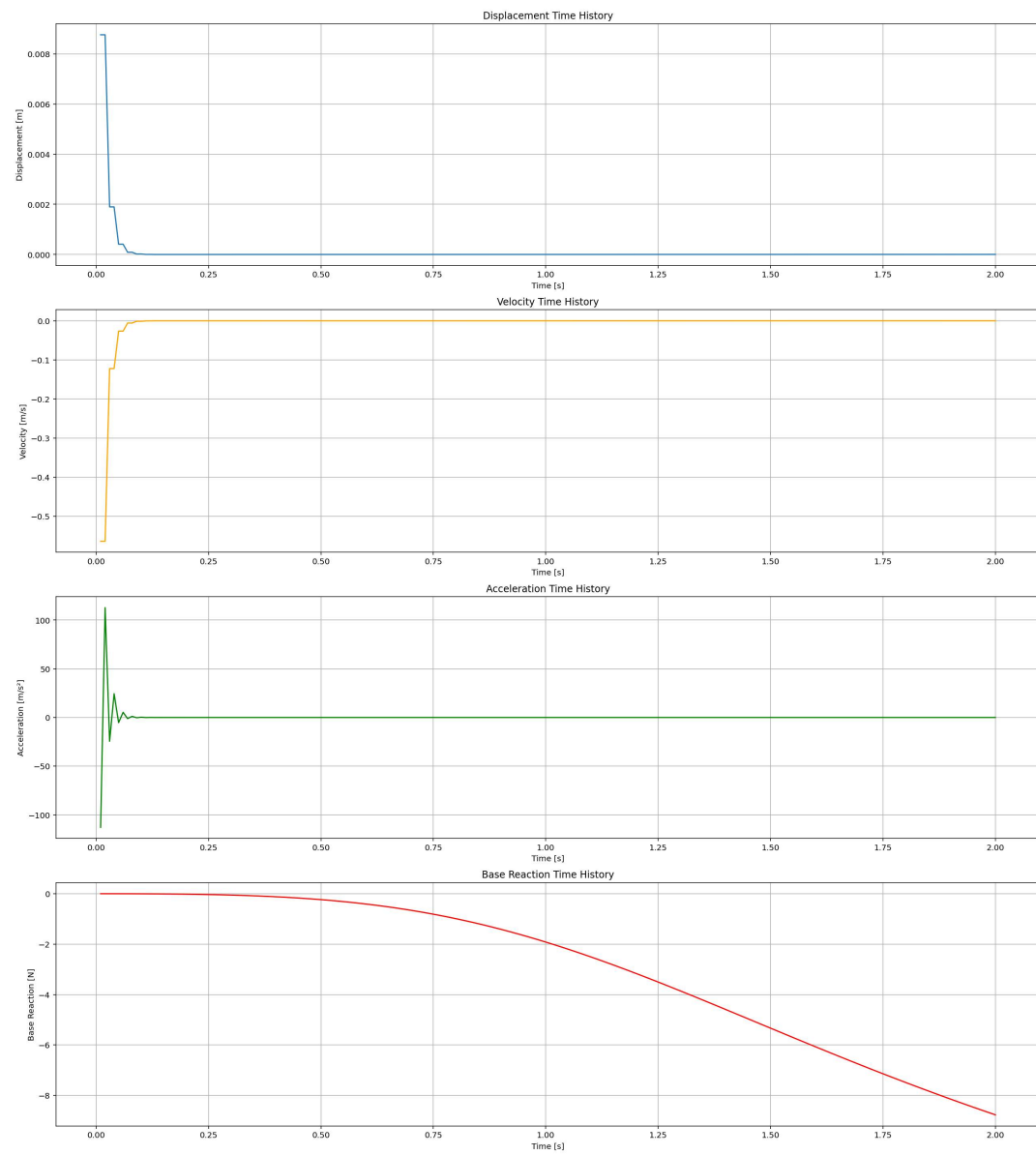












Feature Importance

